

# BSSE FINAL PROJECT

## Design and Test Specification

### Vistelligence



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**Faculty of Information Technology & Computer Science**

**University of Central Punjab**

# **Design and Test Specification**

## **SDP Phase III**

**Vistelligence**

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| Hina Ijaz          | Prototypes + Test specification and<br>results               |
| Kubra Noor         | Test specification and results+<br>Detailed/Component Design |

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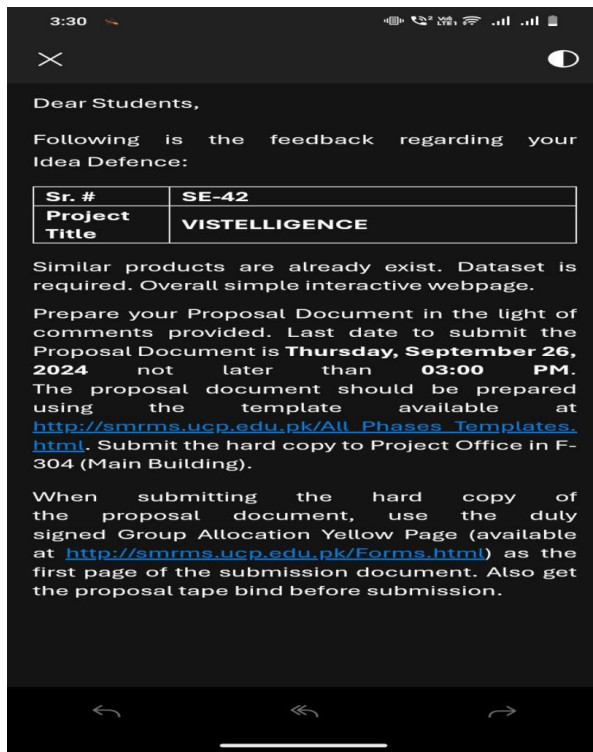
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## Revision History

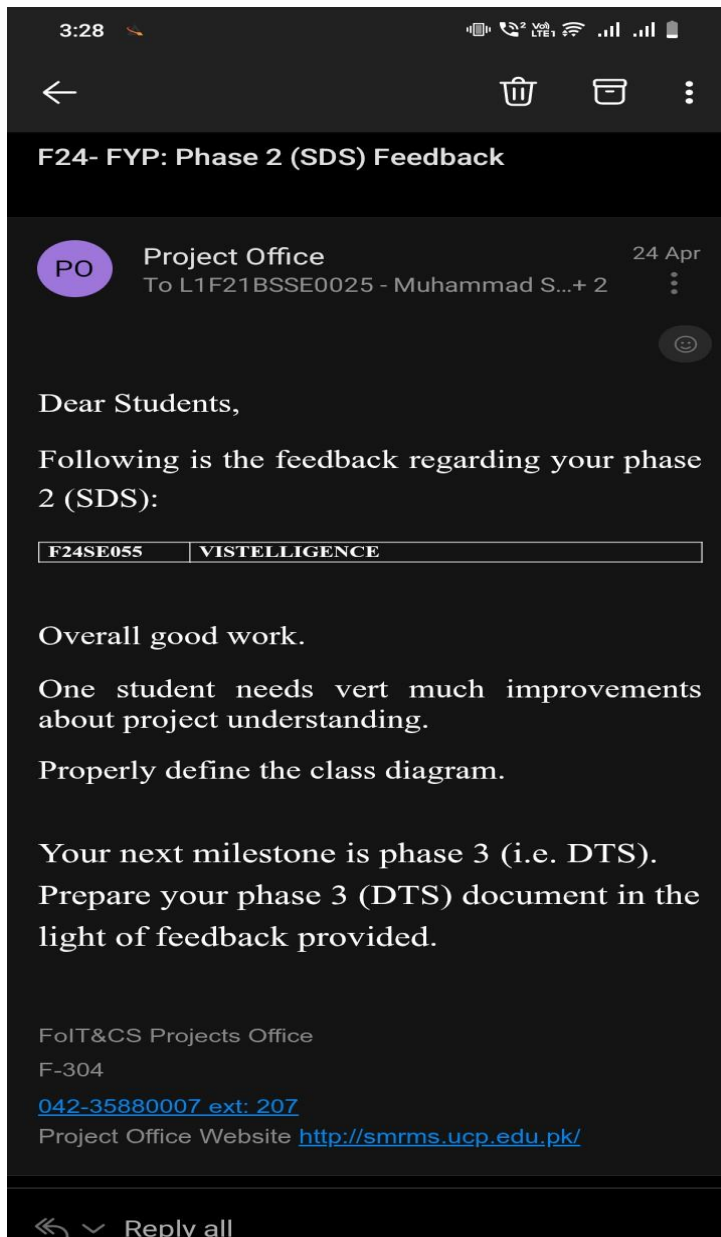
| Name | Date | Reason For Changes | Version |
|------|------|--------------------|---------|
|      |      |                    |         |
|      |      |                    |         |

## Previous Phases Feedback

### Idea Defense Feedback (Screenshot)



### Phase 2 (SDS) Feedback (Screenshot)



## **Abstract**

Vistelligence is an AI-powered web application designed to make the visa application process smoother and more accessible for individuals by analyzing a user's unique profile, including factors like education, financial situation, and career aspirations, to deliver tailored recommendations for visa types, countries, and universities. The platform goes beyond suggestions by predicting visa approval chances, providing interview preparation tools, and offering practical guidance on document preparation, financial planning, and risk management. What sets Vistelligence apart is its focus on enhancing the user experience, with features like a dynamic visa cost estimator to help users budget effectively and an interactive community forum that fosters peer-to-peer support and knowledge sharing, reducing the stress of navigating complex processes while creating a sense of community among users pursuing similar goals. Built on real-time data and advanced AI algorithms, Vistelligence ensures accuracy and reliability in its recommendations, empowering users to make confident decisions about their global mobility journey. Whether seeking to travel, study, or invest abroad, Vistelligence serves as a comprehensive solution, streamlining every step of the process and transforming overwhelming procedures into manageable steps, enabling users to focus on achieving their dreams and building brighter futures on an international stage.

# Introduction

Vistelligence provides personalized visa type recommendations, country and university suggestions, and visa approval predictions. It includes interview preparation, action plans, and real-time updates on policies. The platform features a user-friendly interface and a community forum for advice-sharing. It simplifies the visa process for students, professionals, and investors. Here's the content for each section regarding the document to explain:

## **1.1 Product**

Visa applicants often struggle with understanding which visa types suit their needs, predicting approval chances, and preparing the necessary documentation. The existing platforms provide generic advice, leaving applicants without personalized guidance, leading to rejections and inefficiencies. Vistelligence addresses these issues with AI-powered solutions for personalized visa recommendations and end-to-end process support.

## **1.2 Background**

With the increasing demand for international travel, education, and work opportunities, visa application processes have become a crucial aspect of global mobility. As we know that the current tools lack personalization and rely heavily on static information. Integrating artificial intelligence into this domain can transform the experience, reducing uncertainty and improving approval rates.

## **1.3 Objective(s)/Aim(s)/Target(s)**

The objectives of this platform are:

- It simplifies visa application processes using AI.
- It provides personalized, actionable recommendations for visas, countries, and universities.
- It enhances the success rate of applications with data-driven insights.
- It provides a community for peer-to-peer learning and expert support.

## **1.4 Scope**

Vistelligence covers the entire visa application life-cycle:

- Personalized visa, country, and university suggestions.
- AI-based approval predictions and interview preparation.
- Comprehensive application strategies and real-time cost estimations.
- Community forums for shared experiences and collaborative learning.

This platform caters to students, professionals, and investors seeking visas worldwide. Vistelligence offers an AI-driven platform that simplifies the visa application process, providing personalized recommendations for visa types, universities, and countries. It builds trust by focusing on high success rates and personalized strategies for global mobility.

## 1.5 Business Goals

Business goals of Vistelligence are given below:

- Streamline the visa application process by offering AI-powered guidance and reducing confusion for applicants.
- Increase visa approval chances through personalized recommendations based on user profiles and real-time data.
- Establish a one-stop platform for country, visa, and university selection, saving time and effort.
- Support informed decision-making with accurate cost estimation and strategic planning.
- Build a trusted digital brand in the global education and immigration space through smart tech and community support.

## 1.6 Document Conventions

This Software Requirements Specification (SRS) follows the typographical and formatting conventions outlined below for clarity and consistency:

- **Font Style:** The font used throughout the document is Times New Roman.
- **Font Size:**
  - I. The **body text** is set at **12-point size** to ensure readability.
  - II. **Subheadings** are styled with a **16-point size** for better differentiation and hierarchy.
  - III. **Major page titles** or **section headers** are formatted with a **28-point size** to highlight important sections and maintain a clear structure.
- **Bold Text:** Bold font is used to highlight **key terms** or **important sections** for quick reference.
- **Bullet Points and Numbering:** Bullets and numbered lists are used for clarity and ease of reference in sectioned or grouped information.
- In **Reference section** the name of articles or books are in italic.

## 1.7 Miscellaneous

### ***Team Communication:***

Weekly Zoom meetings and a dedicated Whats App group will be used for updates and discussions to maintain coordination.

### ***Version Control:***

GitHub will be utilized for collaborative development and version management, ensuring seamless teamwork.

### ***Documentation Standards:***

All documentation will follow IEEE standards to maintain clarity and consistency throughout the project.

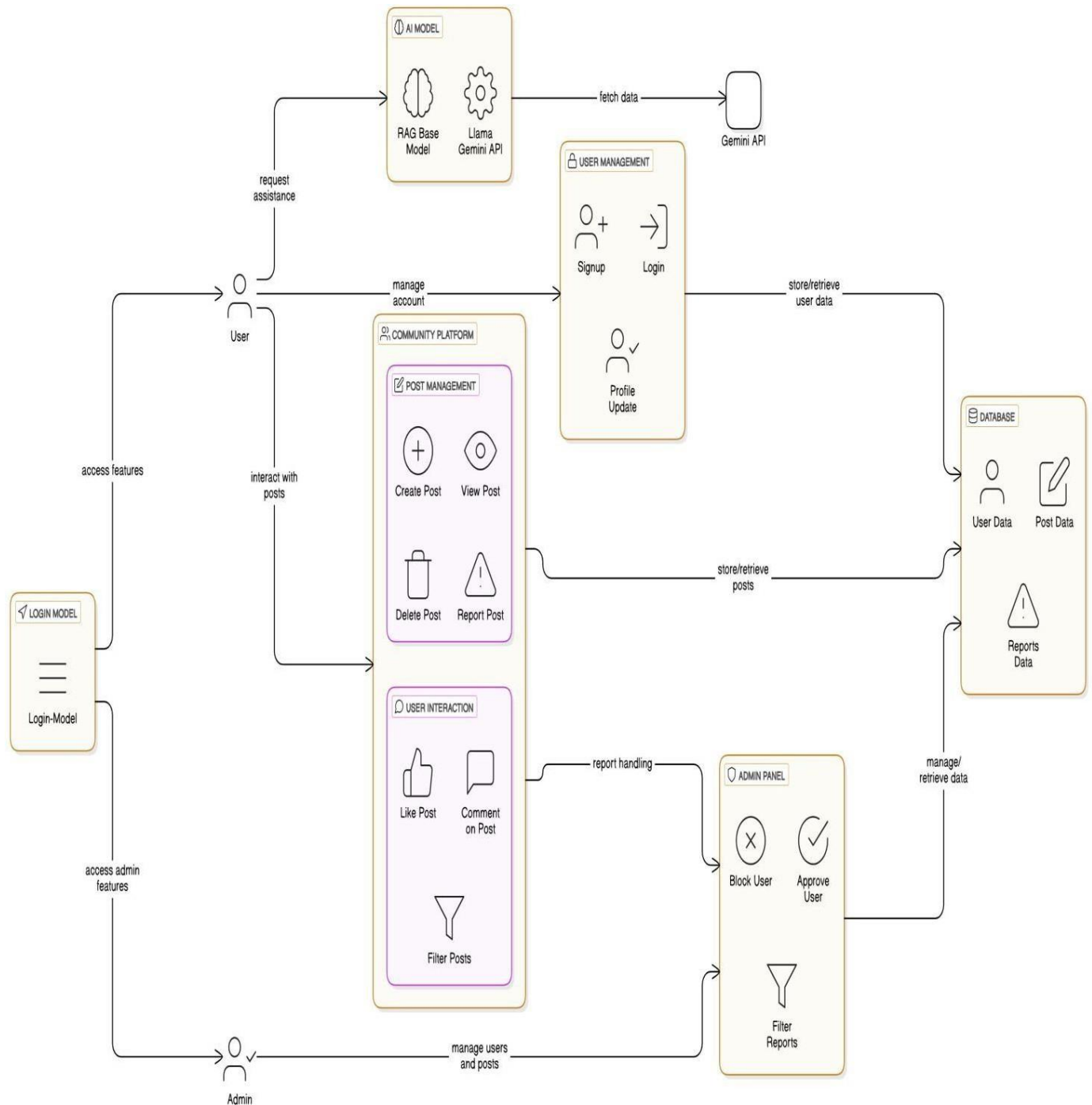


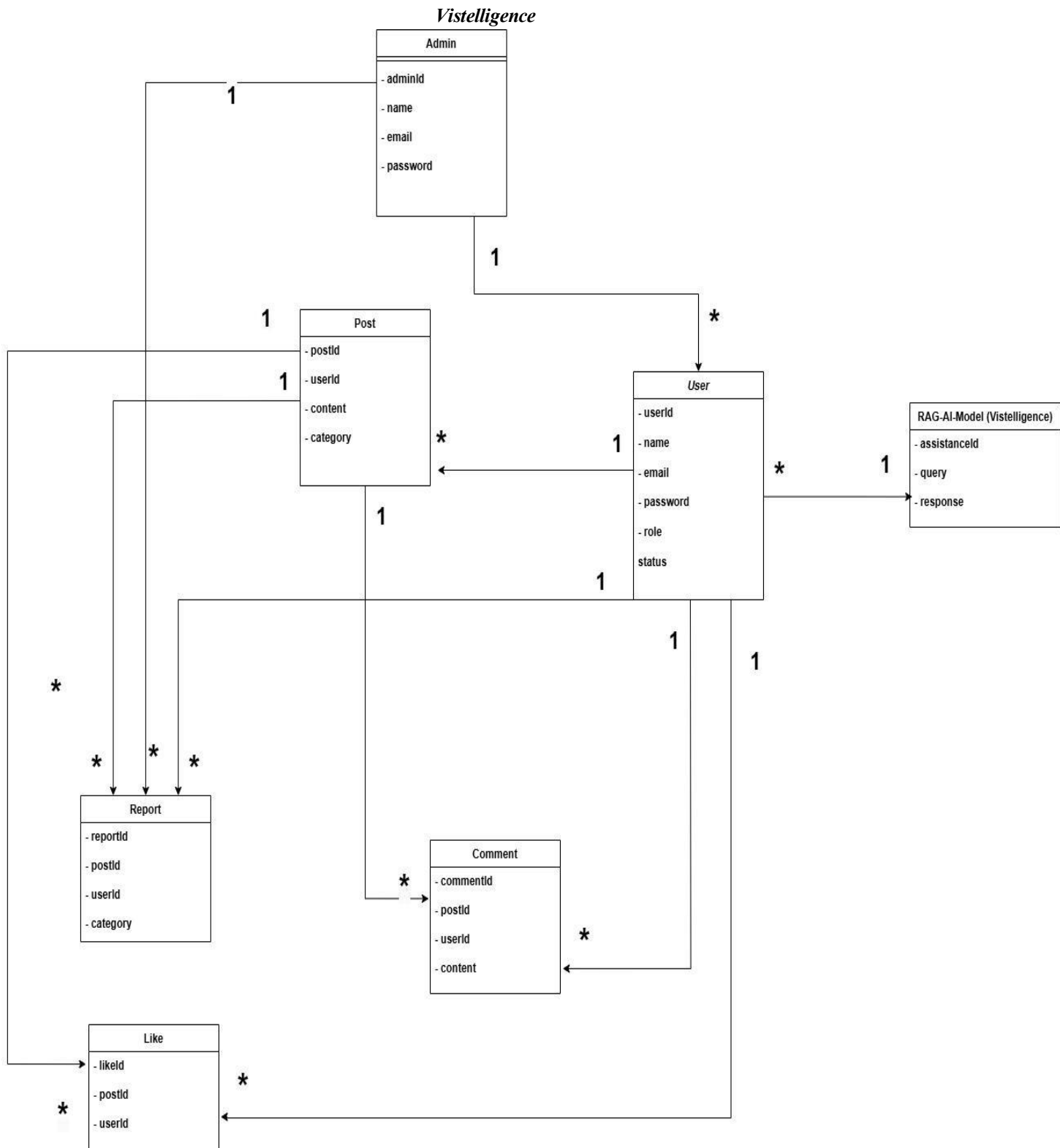
***Design Tools:***

Canva will be used for UI/UX mockups, and Draw.io for creating diagrams and visual models.

## 2. Technical Architecture

### 2.1 Application and Data Architecture





**Figure 2.1.1: ERD diagram:**

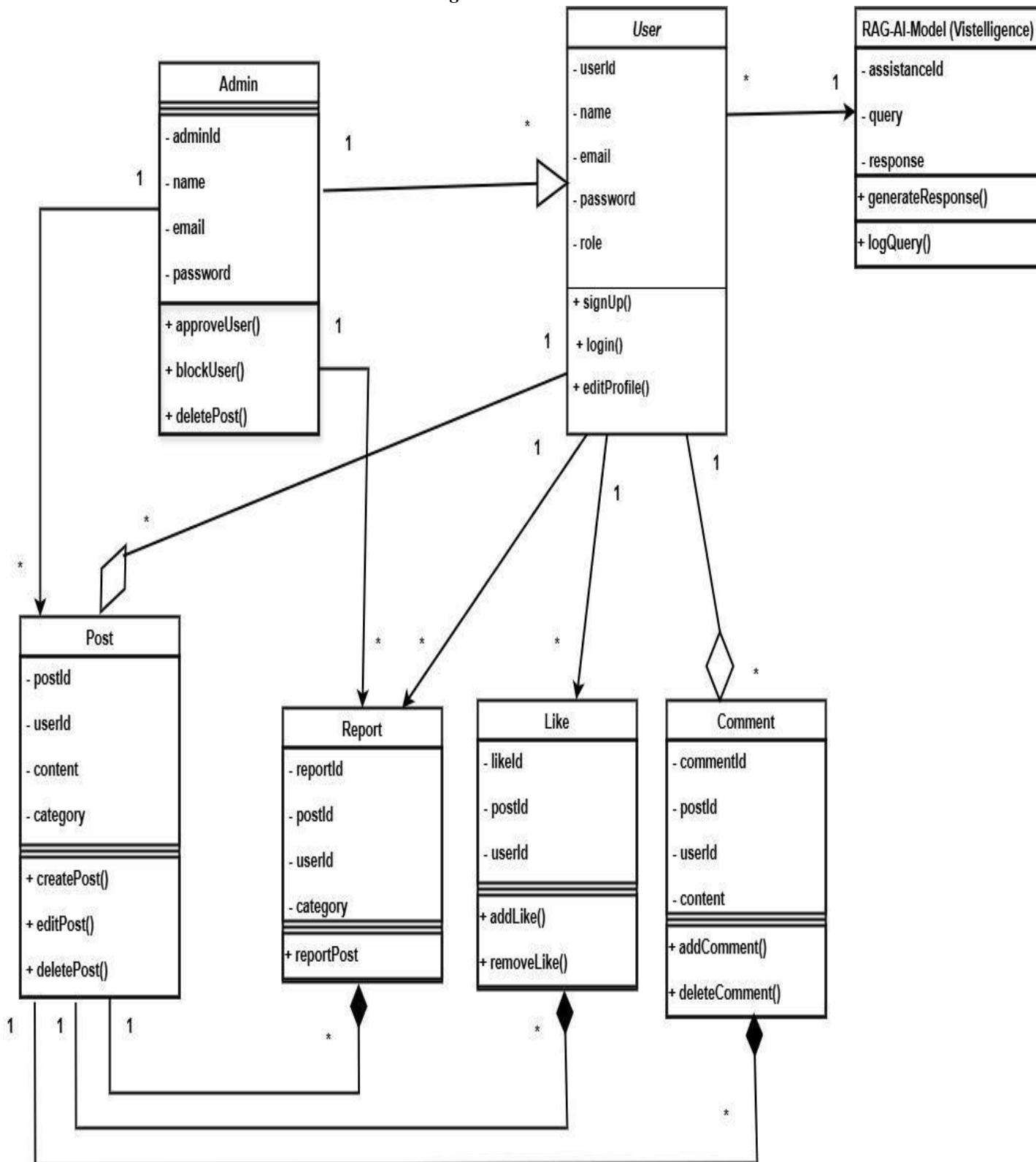
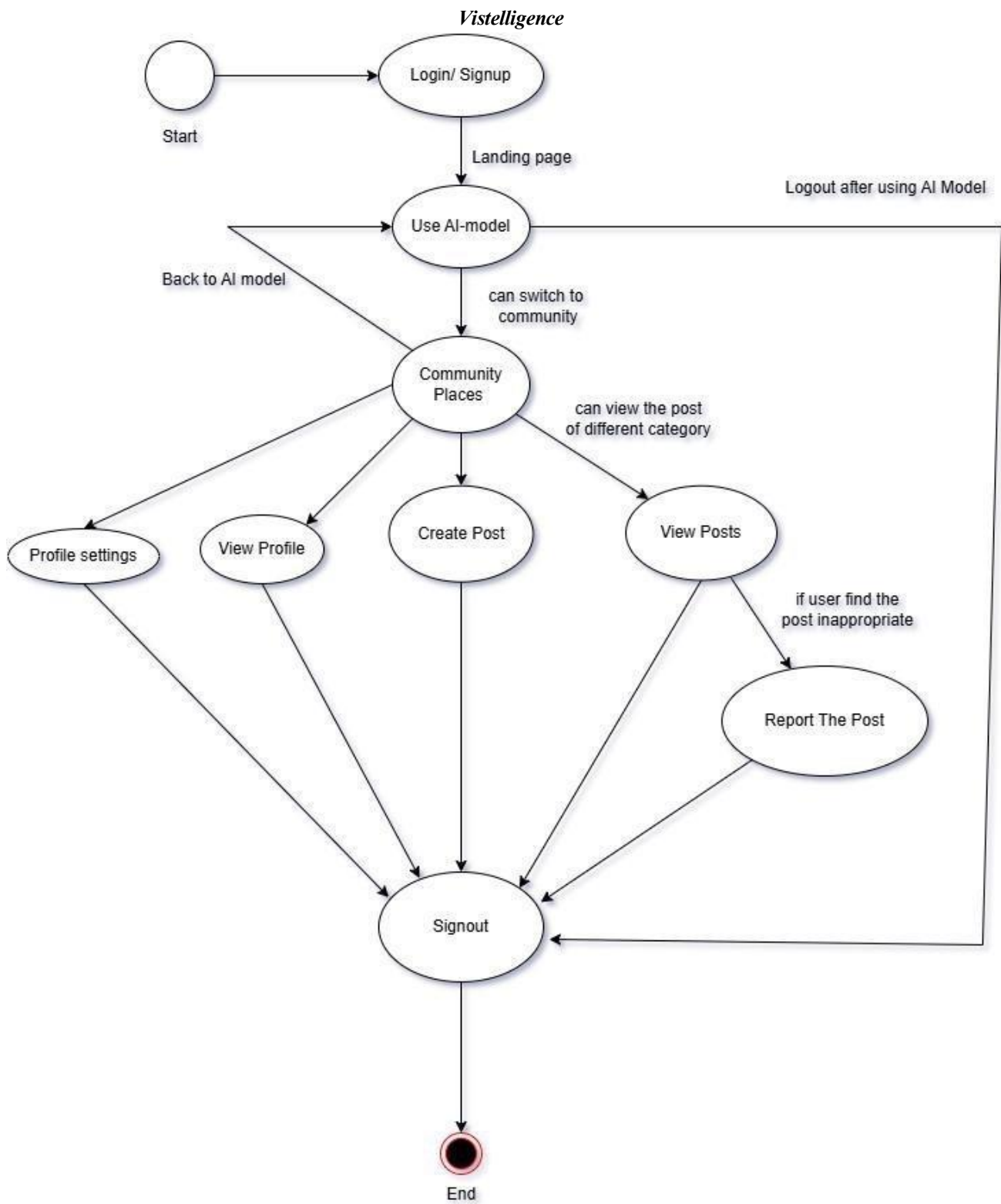


Figure 2.1.2: Class diagram



**Figure 2.1.3: Activity diagram**

**Decision Tables:****Login/Signup Process:**

| Condition | Is Email Valid? | Is Password Strong? | Is Username Unique? | Action                                    |
|-----------|-----------------|---------------------|---------------------|-------------------------------------------|
| Case 1    | Yes             | Yes                 | Yes                 | Register user                             |
| Case 2    | Yes             | Yes                 | No                  | Prompt "Username already taken"           |
| Case 3    | Yes             | No                  | Yes                 | Prompt "Weak password"                    |
| Case 4    | Yes             | No                  | No                  | Prompt "Weak password and username taken" |
| Case 5    | No              | Any                 | Any                 | Prompt "Invalid email"                    |

**AI Model Usage**

| Condition | User Logged In? | Model Available? | User Request Valid? | Action                               |
|-----------|-----------------|------------------|---------------------|--------------------------------------|
| Case 1    | Yes             | Yes              | Yes                 | Grant access to AI model             |
| Case 2    | Yes             | Yes              | No                  | Prompt "Invalid request"             |
| Case 3    | Yes             | No               | Any                 | Prompt "Model currently unavailable" |

*Vistelligence*

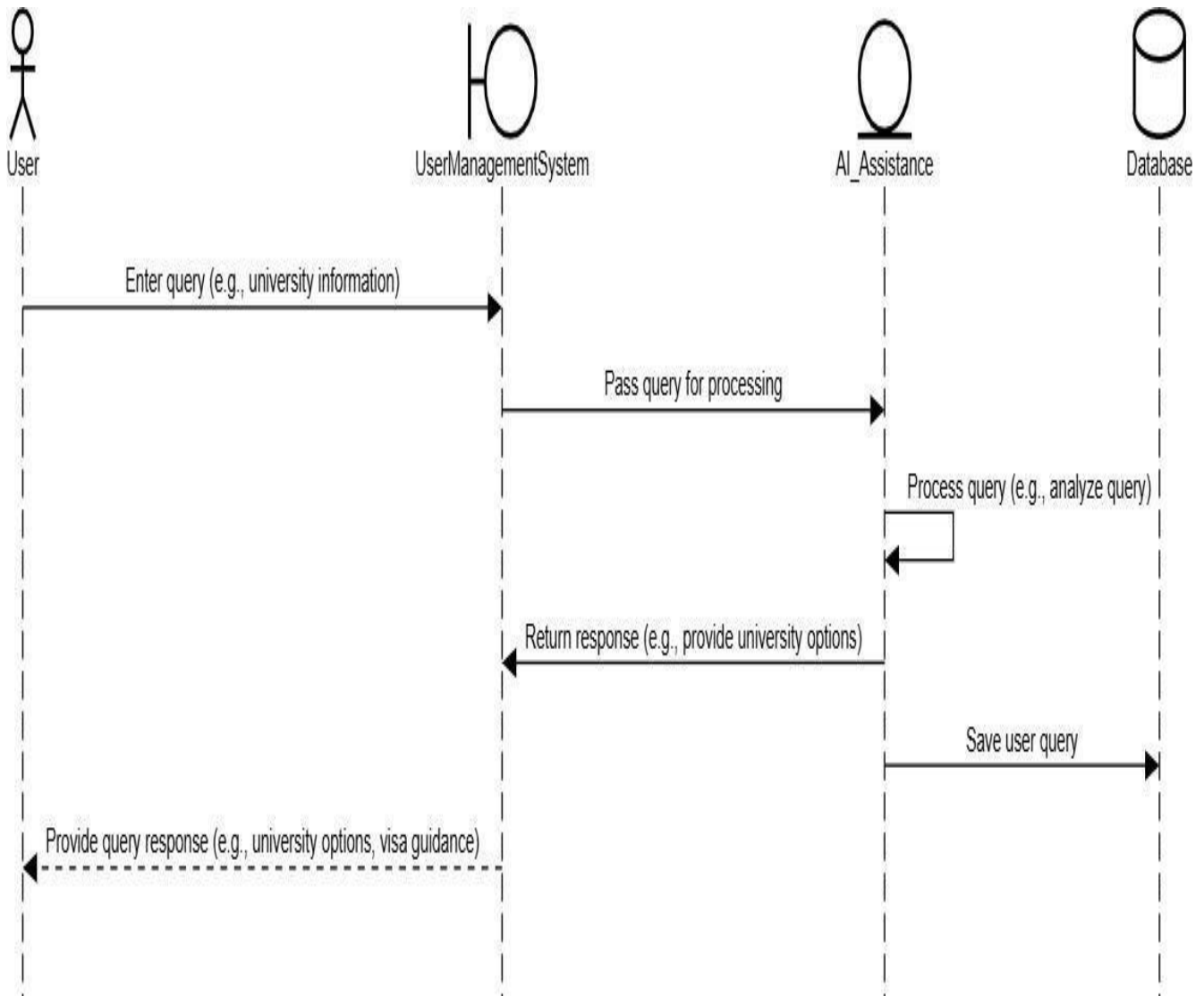
|        |    |     |     |                          |
|--------|----|-----|-----|--------------------------|
| Case 4 | No | Any | Any | Redirect to login/signup |
|--------|----|-----|-----|--------------------------|

Community Places:

| Condition | Post Valid? | Category Available? | User Report Post? | Action        |
|-----------|-------------|---------------------|-------------------|---------------|
| Case 1    | Yes         | Yes                 | No                | Display posts |

|        |     |     |     |                               |
|--------|-----|-----|-----|-------------------------------|
| Case 2 | Yes | Yes | Yes | Mark post as inappropriate    |
| Case 3 | No  | Any | Any | Prompt "Invalid post content" |
| Case 4 | Any | No  | Any | Prompt "Category unavailable" |

## 2.2 Component Interactions and Collaborations



**Figure 2.2.1: AI interaction model**



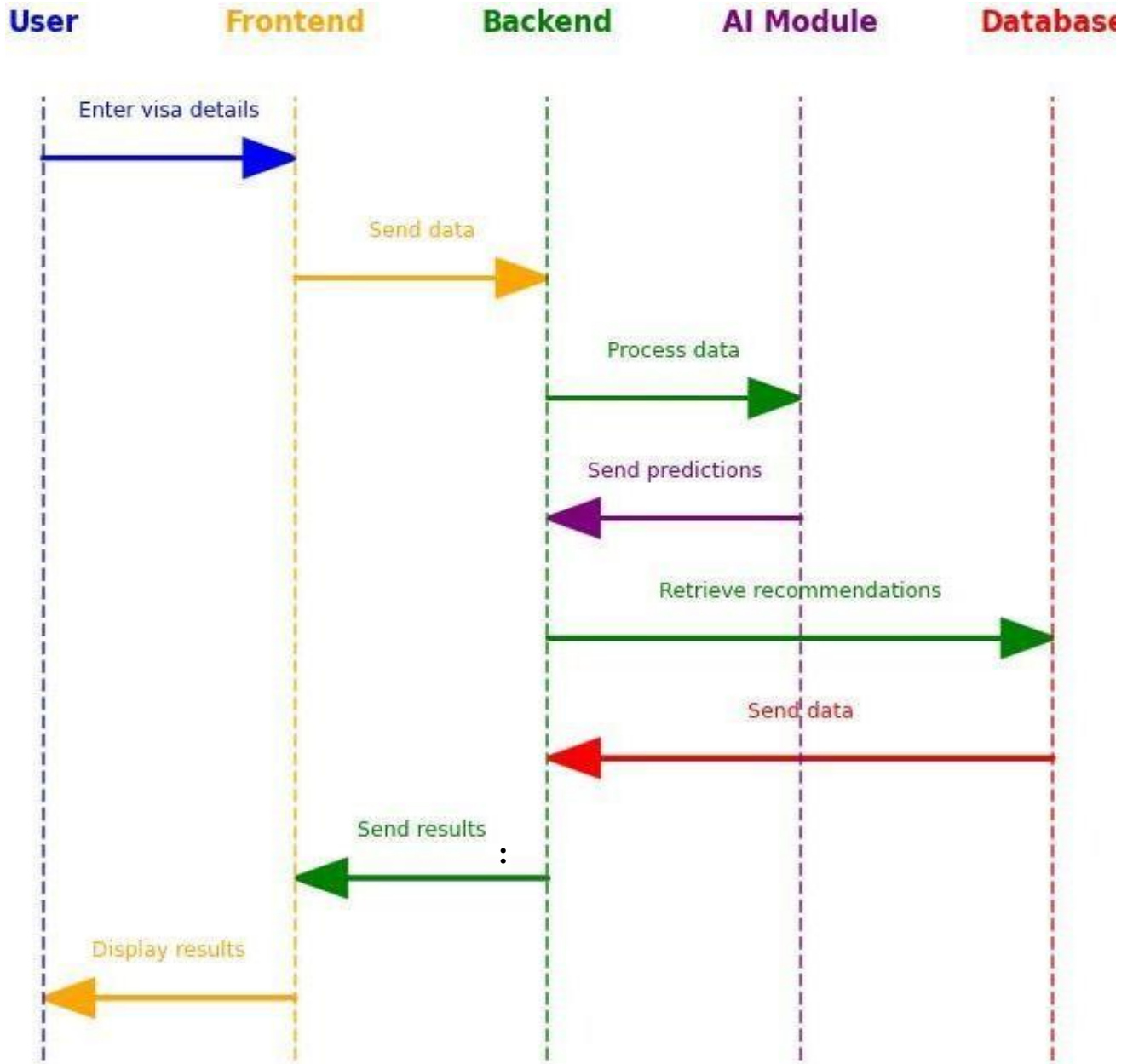


Figure 2.2.2: Event trace diagram

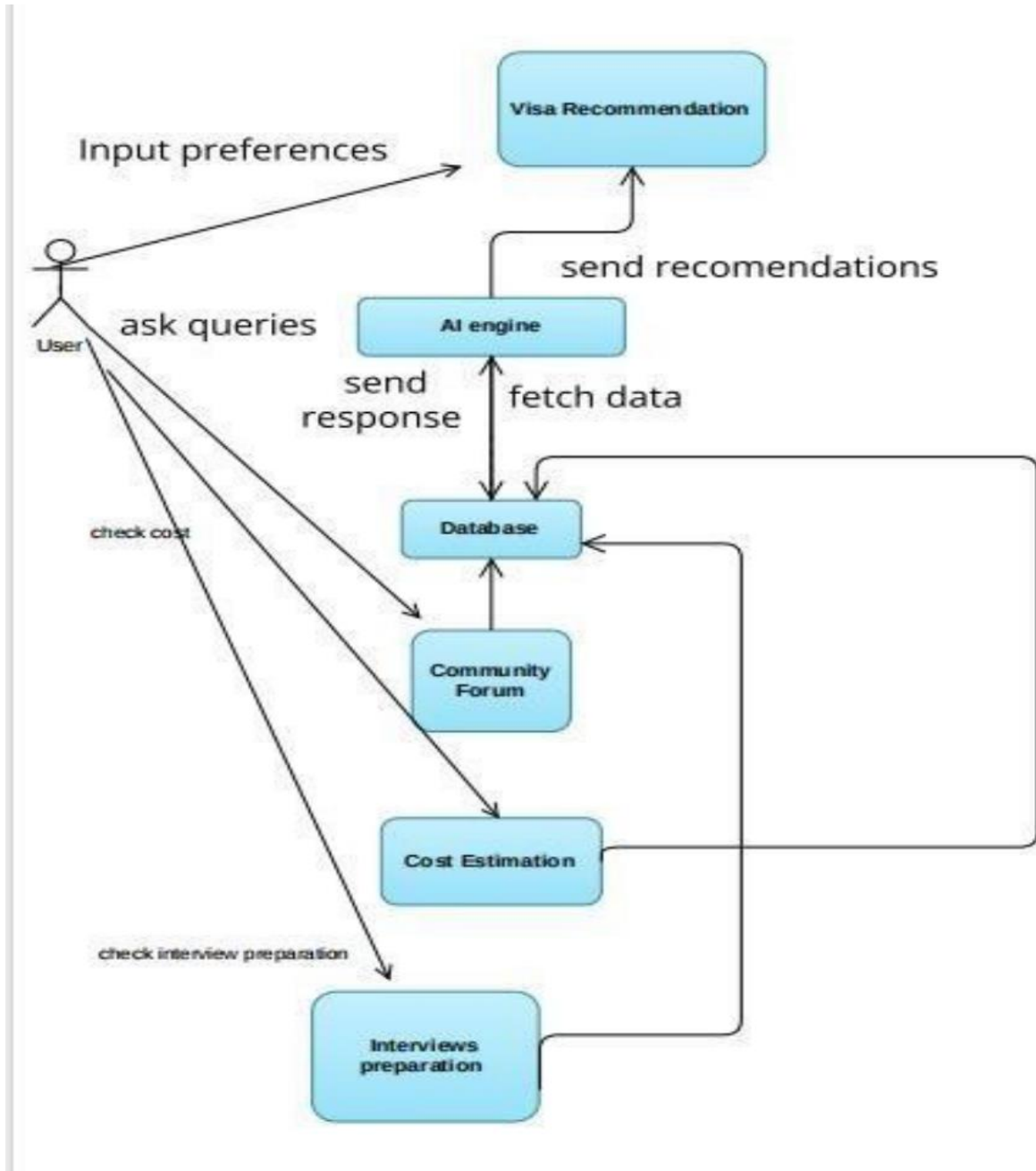


Figure 2.2.3: Collaboration diagram

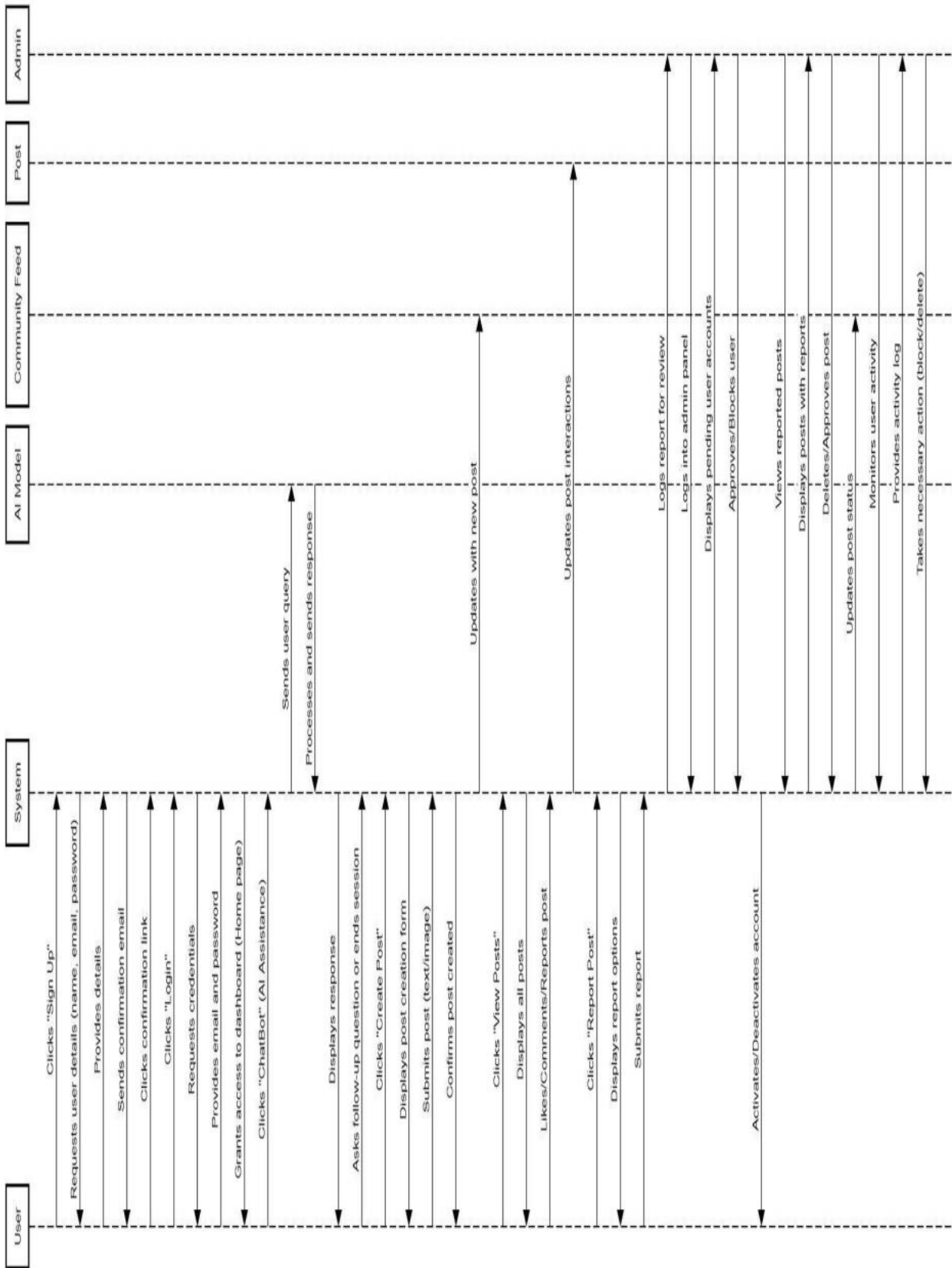


Figure 2.2.4:Swimlane diagram

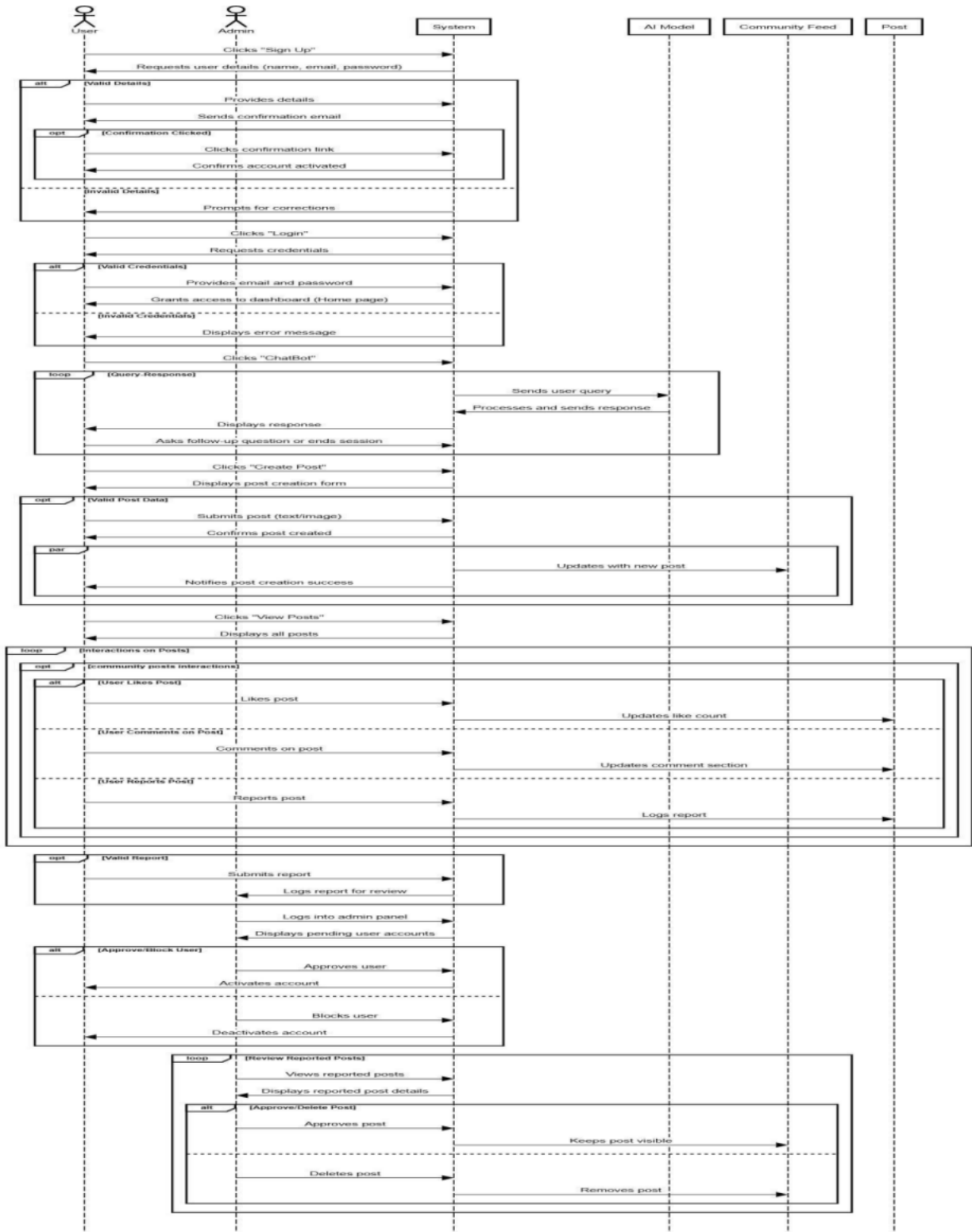


Figure 2.2.5: Sequence diagram

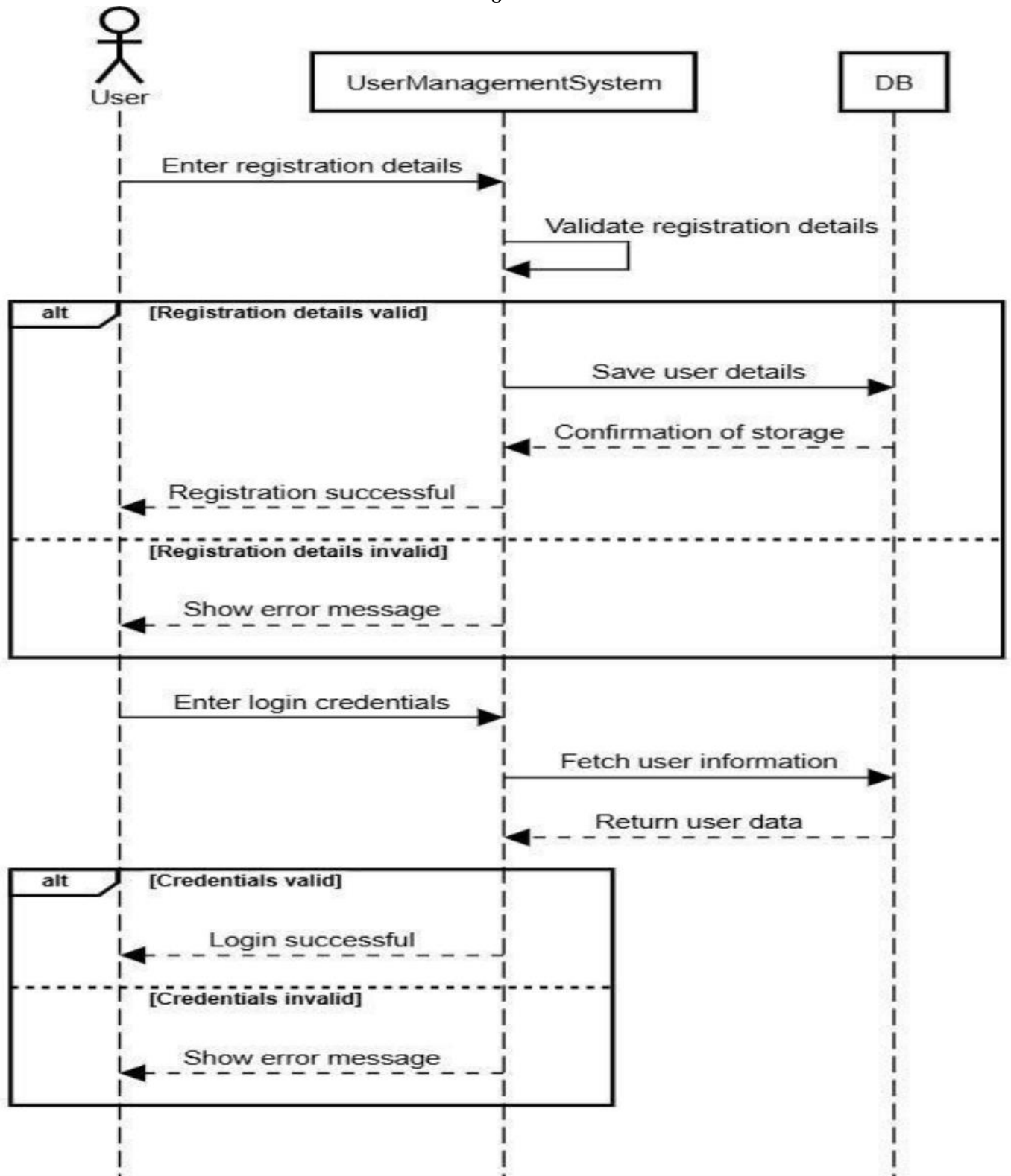


Figure 2.2.6: User Management

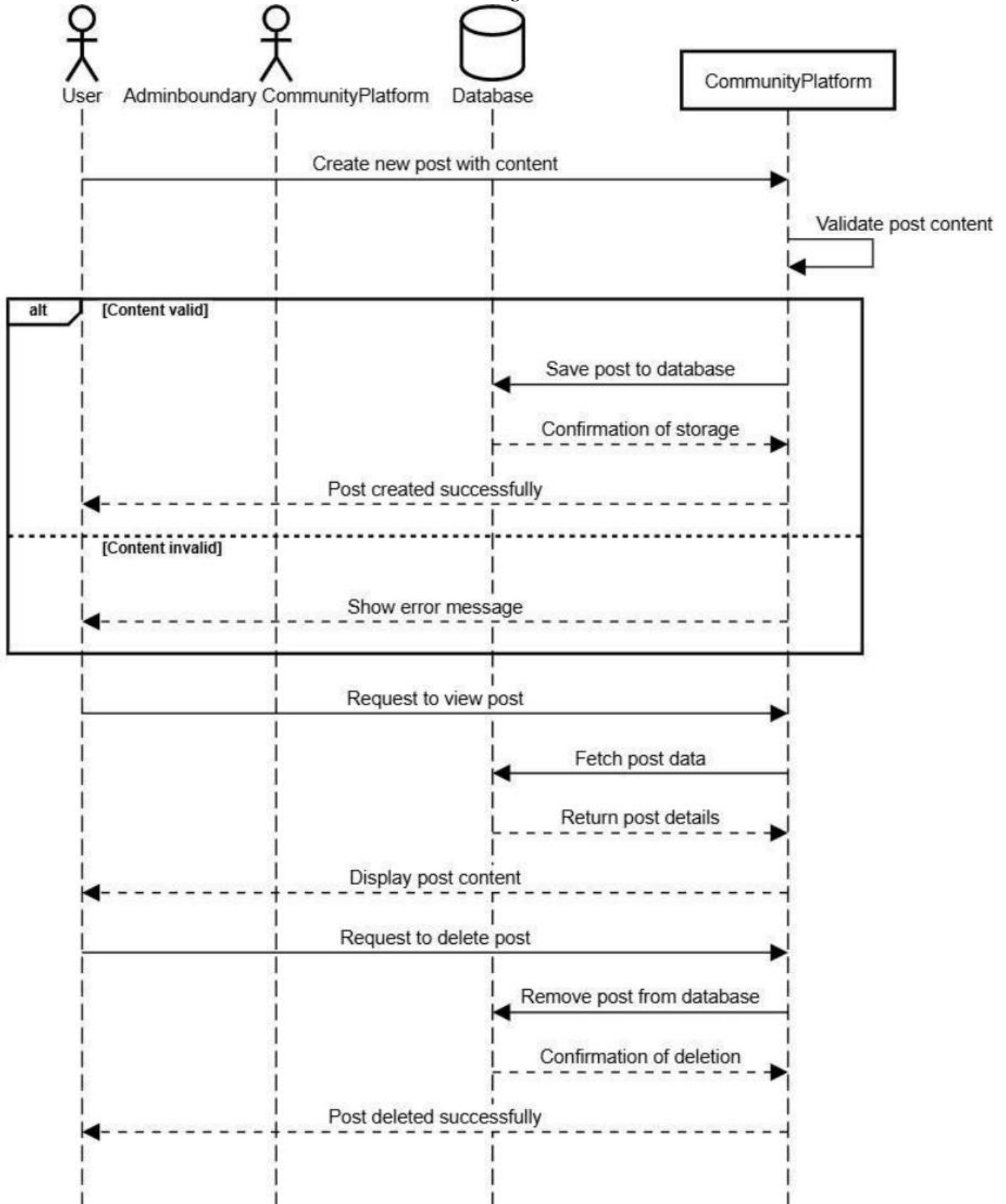


Figure 2.2.7: Creating and Managing post

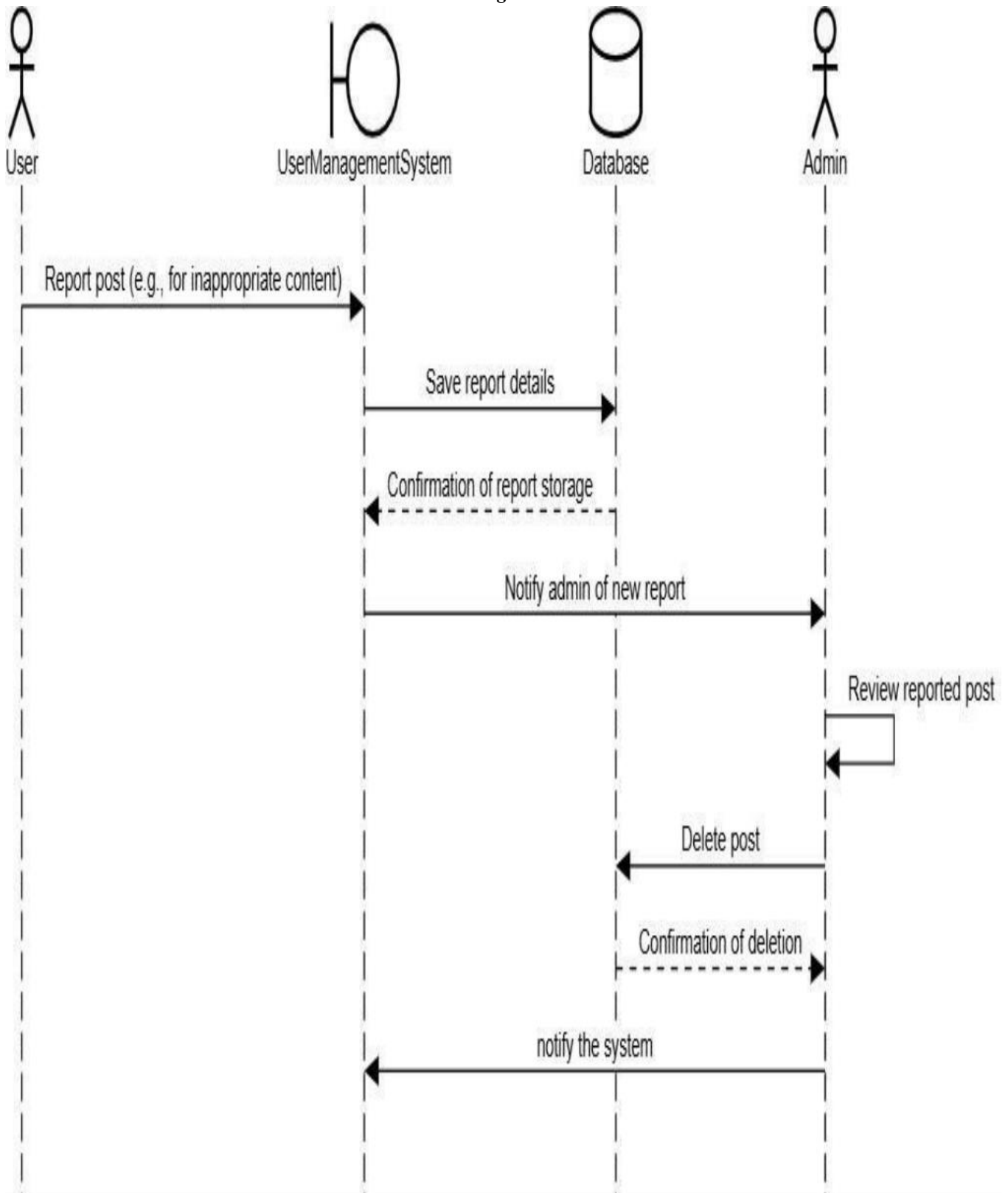


Figure 2.2.8: Report post

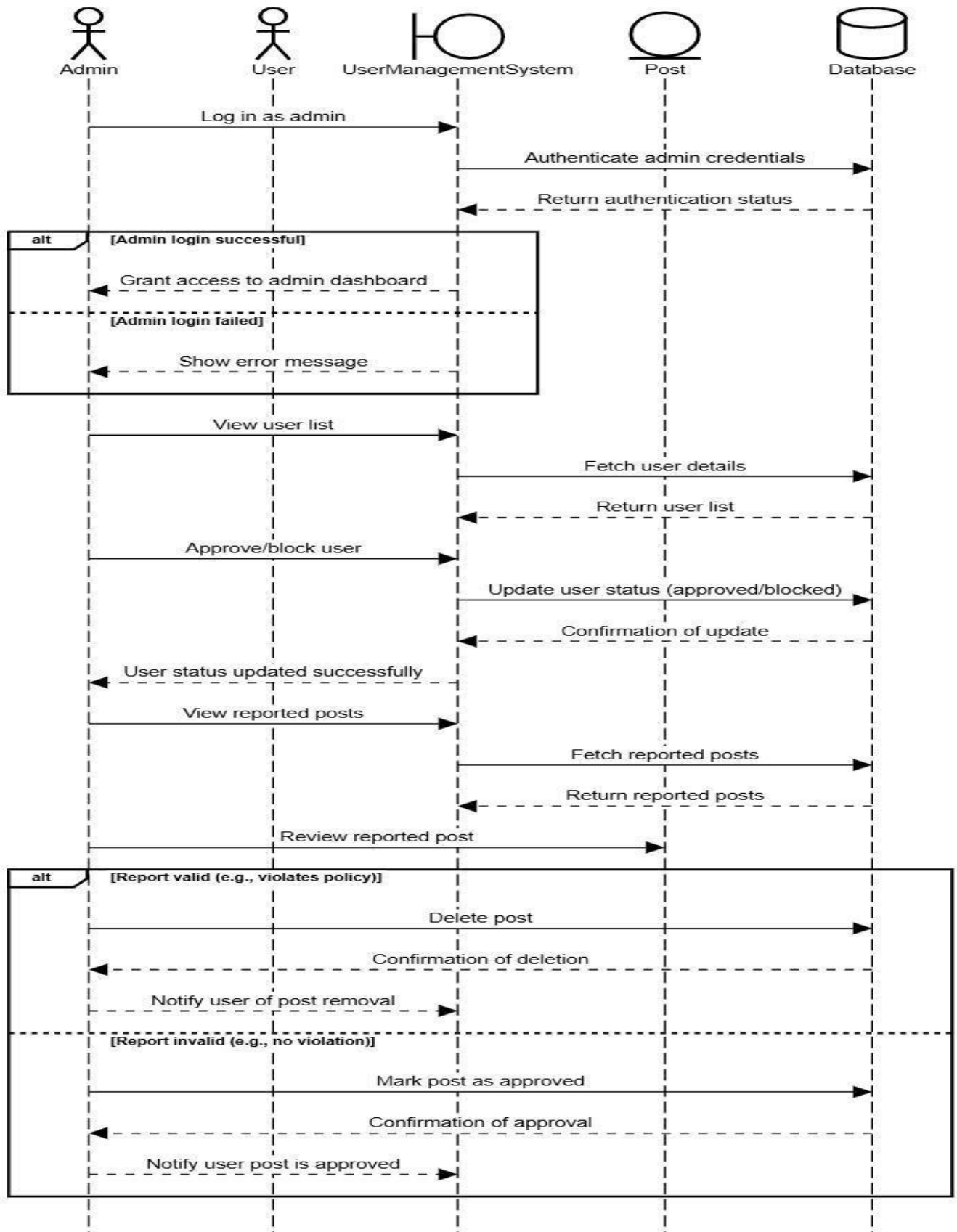


Figure 2.2.9: Admin review and post



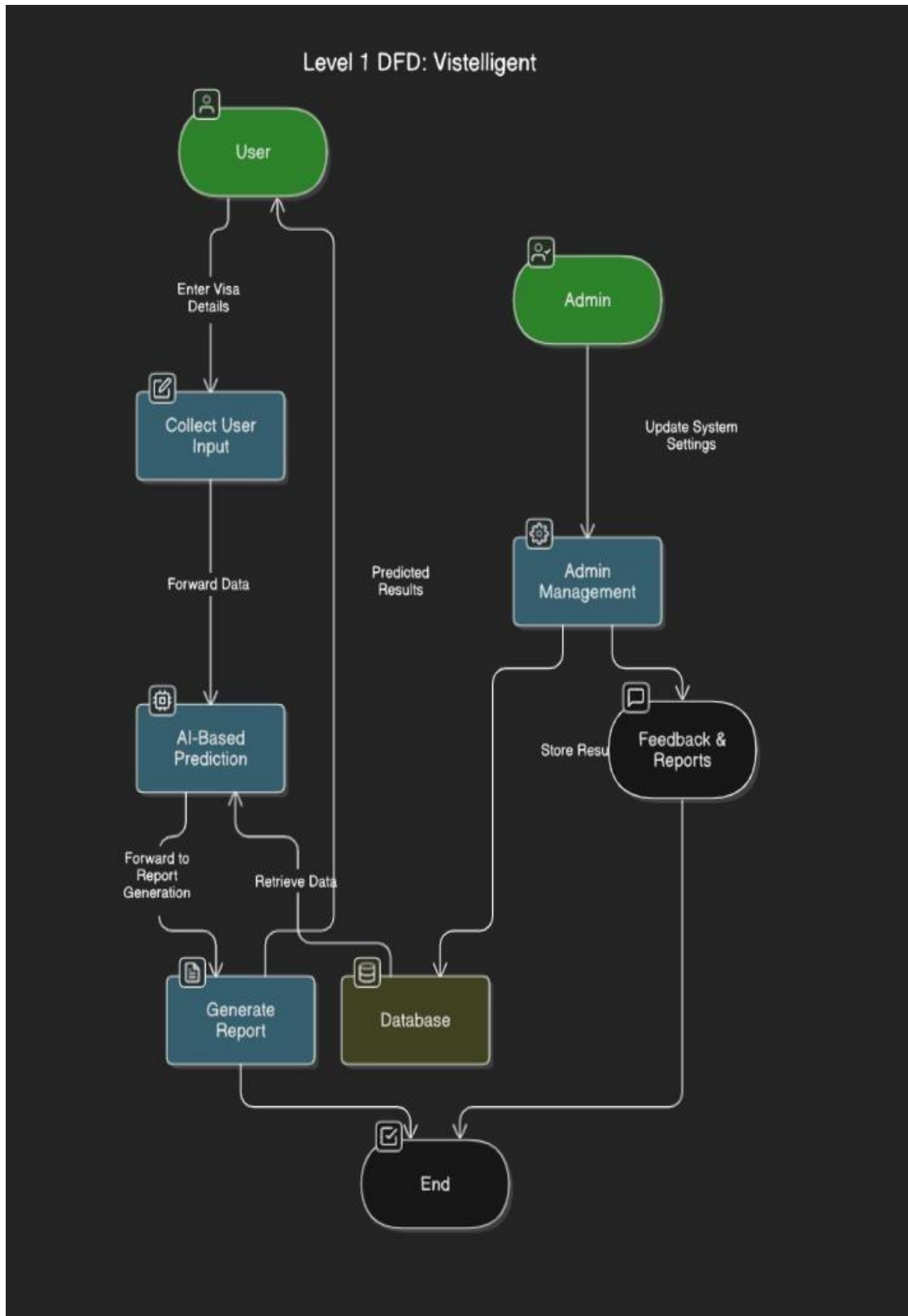


Figure 2.2.10: DFD

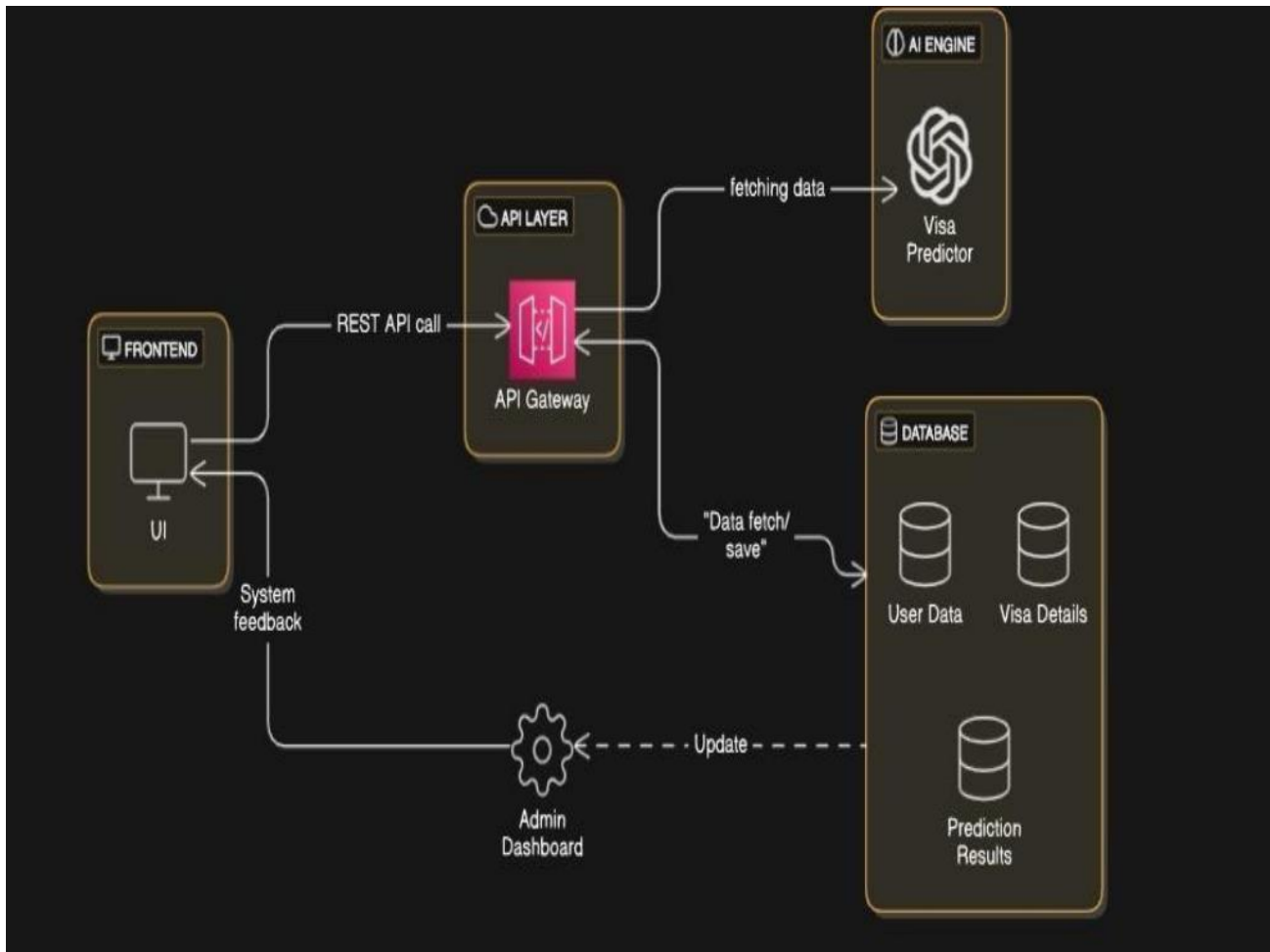


Figure 2.2.11: Component diagram

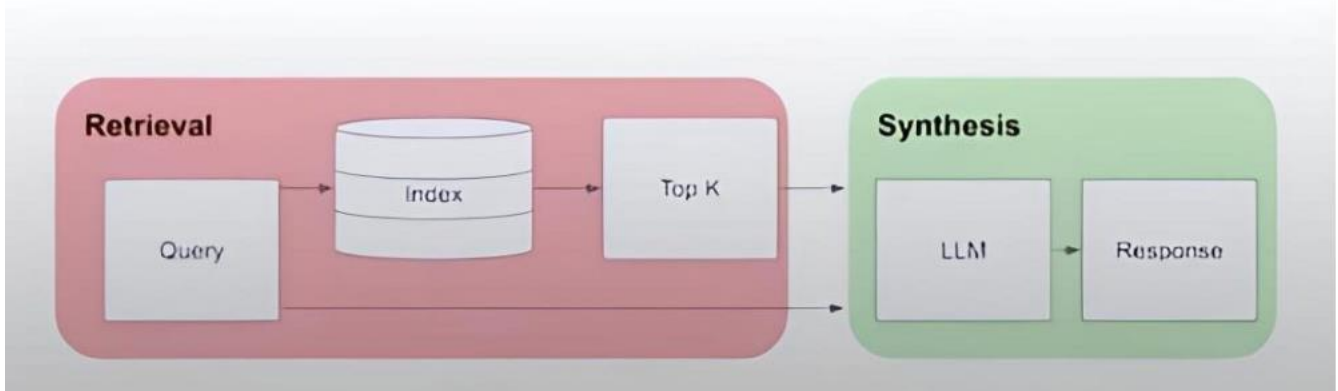
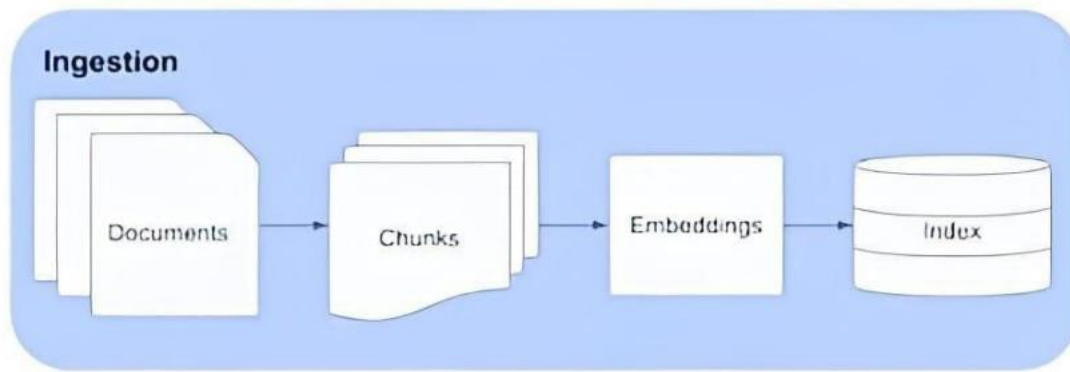


Figure 2.2.12: AI working model

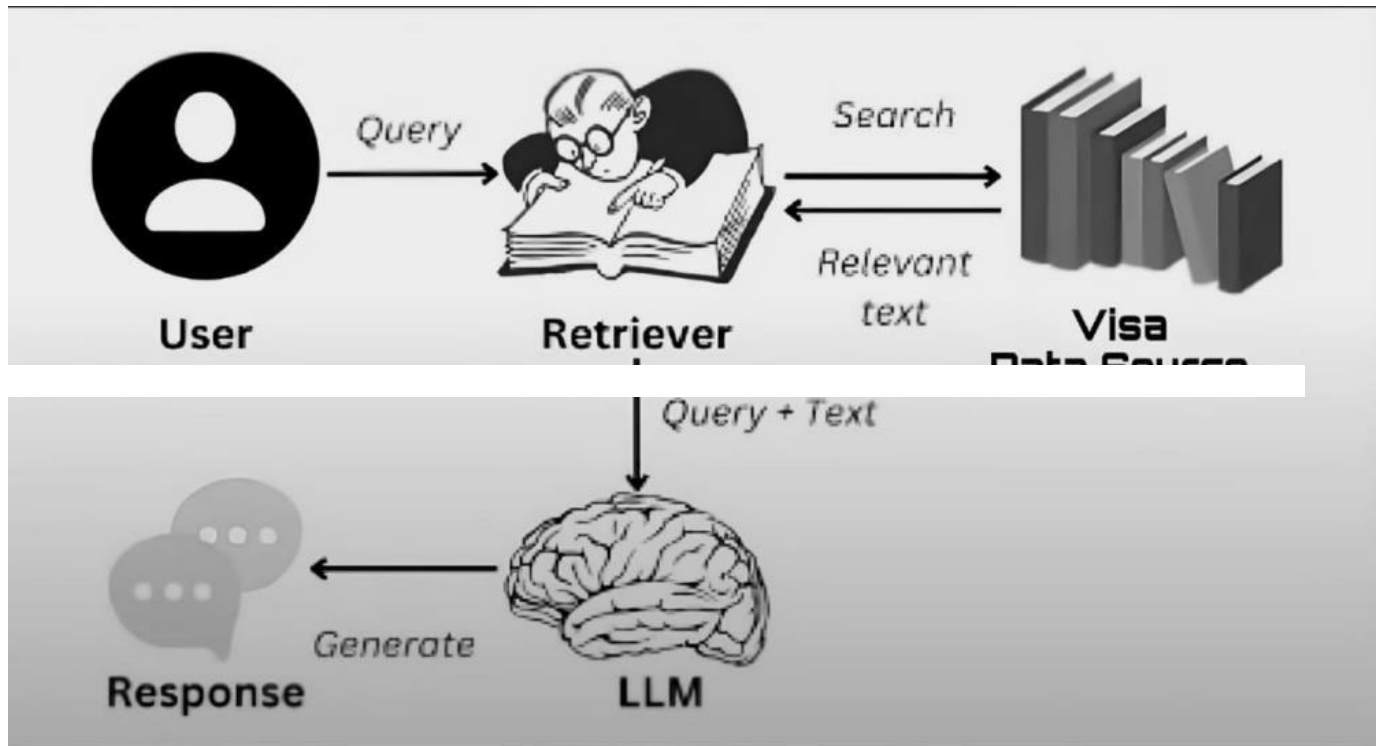
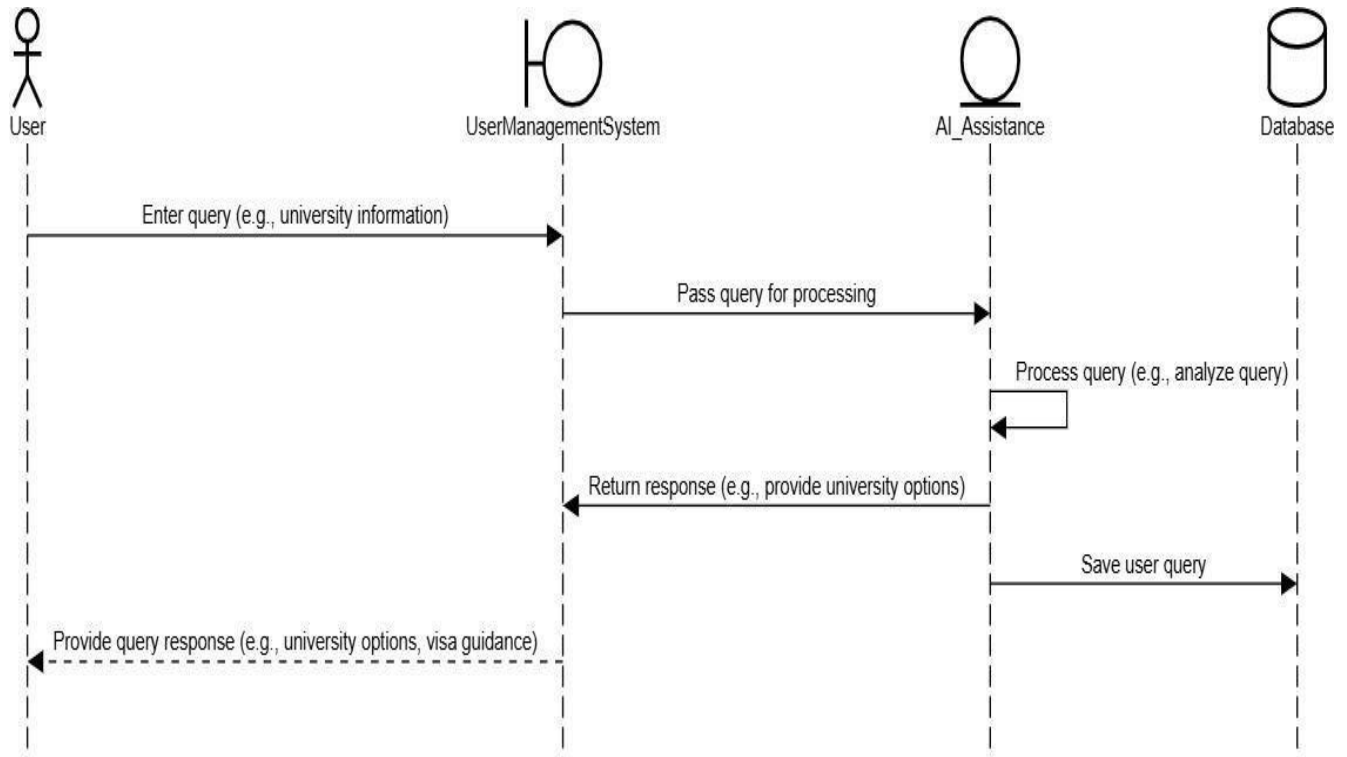


Figure 2.2.13: AI interaction model



## 2.3 Design Reuse and Design Patterns

### Reusable Components:

- **Authentication Module:** A pre-exist, modular authentication system is reused to handle user login, registration, and session management.
- **Payment Gateway Integration:** Reuse of a standard API integration for payment processing, ensuring compatibility and reliability.

### Design Patterns:

- **Singleton Pattern:** Applied in managing configuration settings and logging services to ensure a single, globally accessible instance.
- **Factory Pattern:** Used for creating objects.
- **Observer Pattern:** Implemented for real-time updates, such as notifying users about their queries and suggestions (community forum)
- **User Interface Templates:** Standardized UI components (buttons, forms, and dashboards) from a library like Bootstrap were reused to ensure a consistent and responsive design.

### Database Schemas:

- Predefined schema templates were adapted for managing user data, portfolios, and transactions, reducing the time required for database design.

## 2.4 Technology Architecture

### Platform

- **Front-end:** Built using HTML, CSS, and JavaScript frameworks like React for a responsive and user-friendly interface.
- **Back-end:** Powered by Node.JS to handle server-side tasks and manage APIs.

### System Hosting

- Hosted on cloud platforms like AWS or Google Cloud for reliability and scalability.
- Databases such as MySQL or MongoDB store user and transaction data.

### Connectivity Requirements

- Requires stable internet for user interactions and data syncing.
- Uses HTTPS for secure communication and integrates with third-party APIs like payment processors.

### Modes of Operation Online Mode:

- Two-factor authentication (2FA) and encryption protect user data.
- Regular backups and recovery plans ensure safety.

### Network

Globally accessible over the internet with secure HTTPS connections.

## 2.5 Architecture Evaluation

### Why These Choices?

- **React vs. Angular/Vue:**  
We chose React because it's easy to build with and has a huge community. It's great for making changes quickly and keeping our code organized. Our team also really likes how data flows in one direction and how hooks make things easier.
- **Node.JS vs. Django/Rails:**  
Node lets us use JavaScript for both the front end and back end, which means less switching between languages. It's fast and works well for what we need.
- **AWS vs. On-Prem/Other Clouds:**  
AWS gave us the tools we needed, plus it's easy to grow with. We only pay for what we use, and it works all over the world.
- **MySQL vs. PostgreSQL/MongoDB:**  
We use MySQL for important user data because it's reliable and structured. For things like logs and analytics, we also use MongoDB it's better for storing flexible data.

### Pros & Cons

- **React**  
*Pros:* Fast updates on the screen, lots of community support, and easy-to-reuse building blocks.  
*Cons:* Takes time to set up, and some features can be tricky to learn at first.
- **Node.js**  
*Pros:* Same language (JavaScript) on both front and back ends, handles lots of requests well, and has tons of useful packages.  
*Cons:* Can get messy with too many callbacks, and it runs on one thread (though there are ways to work around that).
- **AWS**

### *Vistelligence*

*Pros:* Easy to scale, available in many places worldwide, and comes with a lot of ready-to-use tools.

*Cons:* Pricing can be confusing, and it's easy to get tied into their system.

- **MySQL + MongoDB**

*Pros:* MySQL is great for structured, reliable data. MongoDB is flexible and good for changing or unstructured data.

*Cons:* Using two databases means more work, and keeping data in sync can be tricky.

### **Alternatives Considered**

- **Front-end:** Vue.js for simplicity; dropped because React's ecosystem better matched our AI-driven widgets.
- **Back-end:** Python/Django for rapid prototyping; passed for Node.js to keep language uniform.
- **Cloud:** Google Cloud's AI suite was tempting; AWS won by offering more mature DevOps tooling.
- **Database:** PostgreSQL's advanced features are on our roadmap; for now, MySQL's stability fits our timeline.

## **3. Detailed/Component Design**

In this section, detailed information regarding the types of interactions and collaborations required within the project is provided.

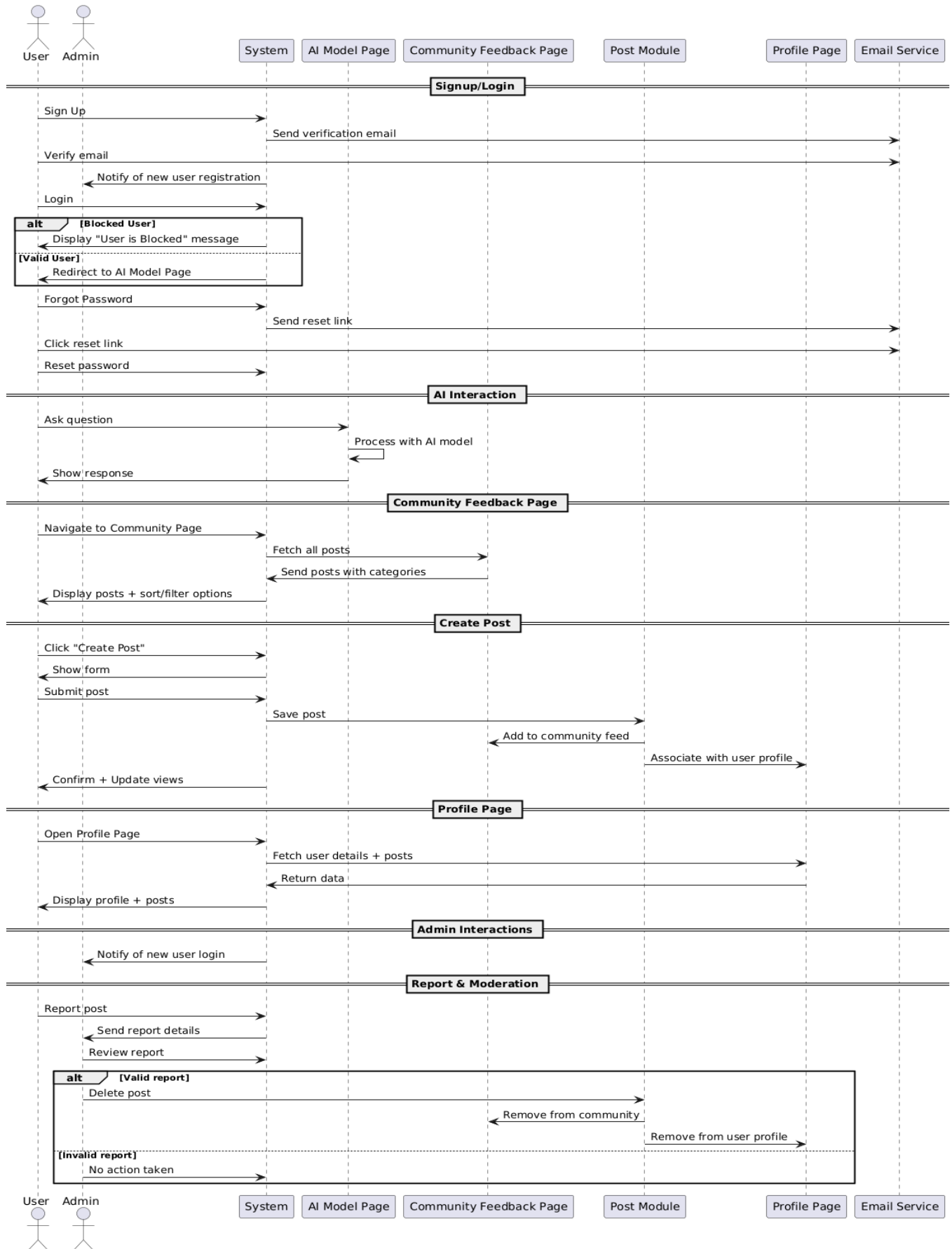
### **3.1 Component-Component Interface**

The system is designed with a modular architecture where the front-end, back-end, and AI modules interact through REST APIs. Key interactions include:

- **User Interface → Authentication Service:** For login, signup, and profile updates.
- **User Interface → Visa Recommendation Engine:** For personalized visa suggestions.
- **Community Module → Post Management Service:** For creating, editing, and reporting posts.
- **Admin Panel → User Management Service:** For blocking/unblocking users and managing posts.

These interactions are managed using API calls, token-based authentication (JWT), and real-time data fetching (Axios, Fetch API).

# Vistelligence





### Figure 3.1.1 : Sequence diagram

For further details, review ; **Figure 3.2.3: Collaboration diagram** and **Figure 2.2.5, to 2.2.9.**

## 3.2 Component-External Entities Interface

The system interacts with the following external services:

- **SMTP Email Server:** For user email verification and password reset (example: AWS SES).
- **Payment Gateway** (planned): Integration with Stripe or PayPal for visa-related service payments.
- **Third-party APIs** (future expansion): For real-time visa policy updates.

Example flow:

- After user signup, the system calls the SMTP server to send a verification email.
- Payment gateway APIs will later be used to securely process payments.

## 3.3 Component-Human Interface

**Screens where users interact with the system:**

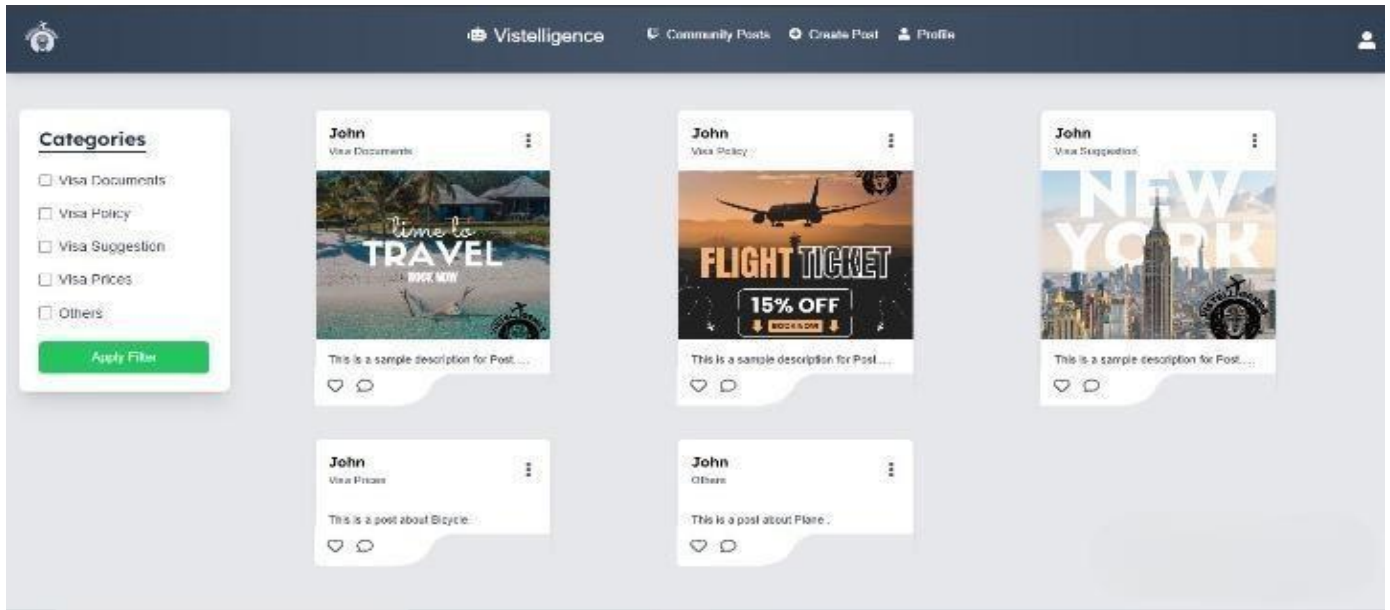
- Registration/Login pages
- Dashboard (Profile, Recommendations, Community Access)
- Post Creation and Community Interaction Pages
- Admin Panel (Manage Users and Posts)

**HCI Norms Followed:**

- Simple, intuitive, and consistent design (React + Bootstrap).
- Proper use of validation messages for user input errors.
- Responsive design for mobile, tablet, and desktop screens.
- Use of icons and colors for accessibility.

## 4. Screenshots/Prototype

### Community Posts:



### Footer:



### Vistelligence:



## Create Post Page:

**Create Post**

Profile Image URL  
Enter profile image URL

Username  
Enter your name

Category  
Select a category

Post Image (jpg, png, pdf)  
Choose File No file chosen

Description  
Write a description

Post

## Setting Page:

**Setting**

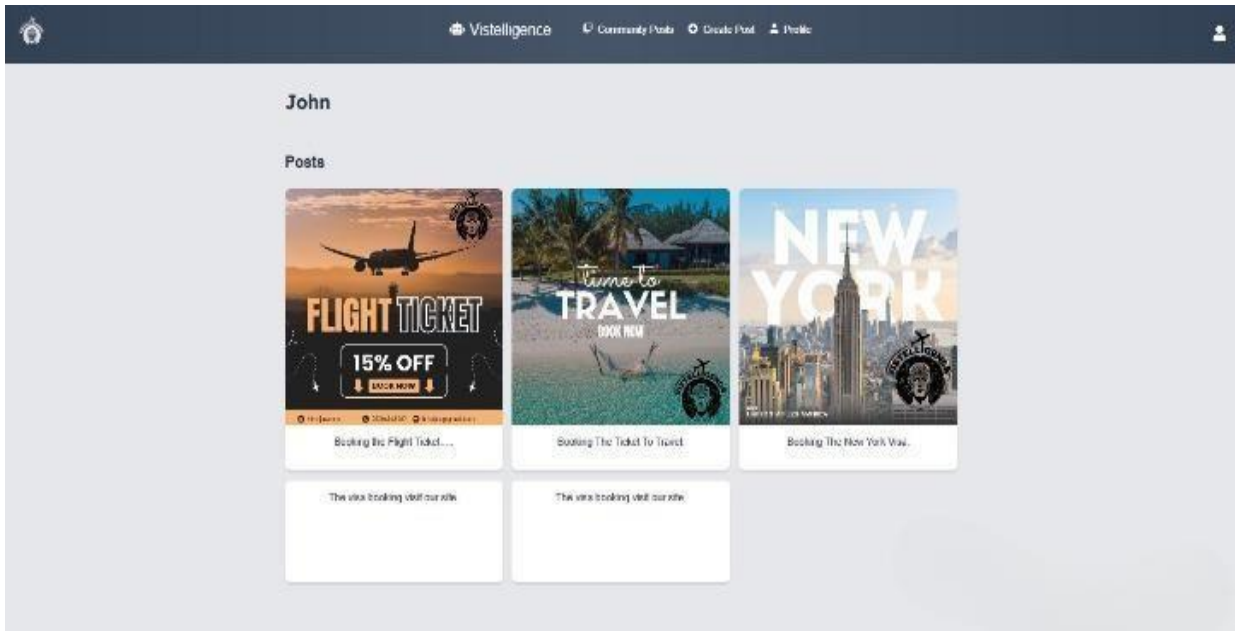
Name  
Abdul Wahab

Email  
wahabnadeem311@gmail.com

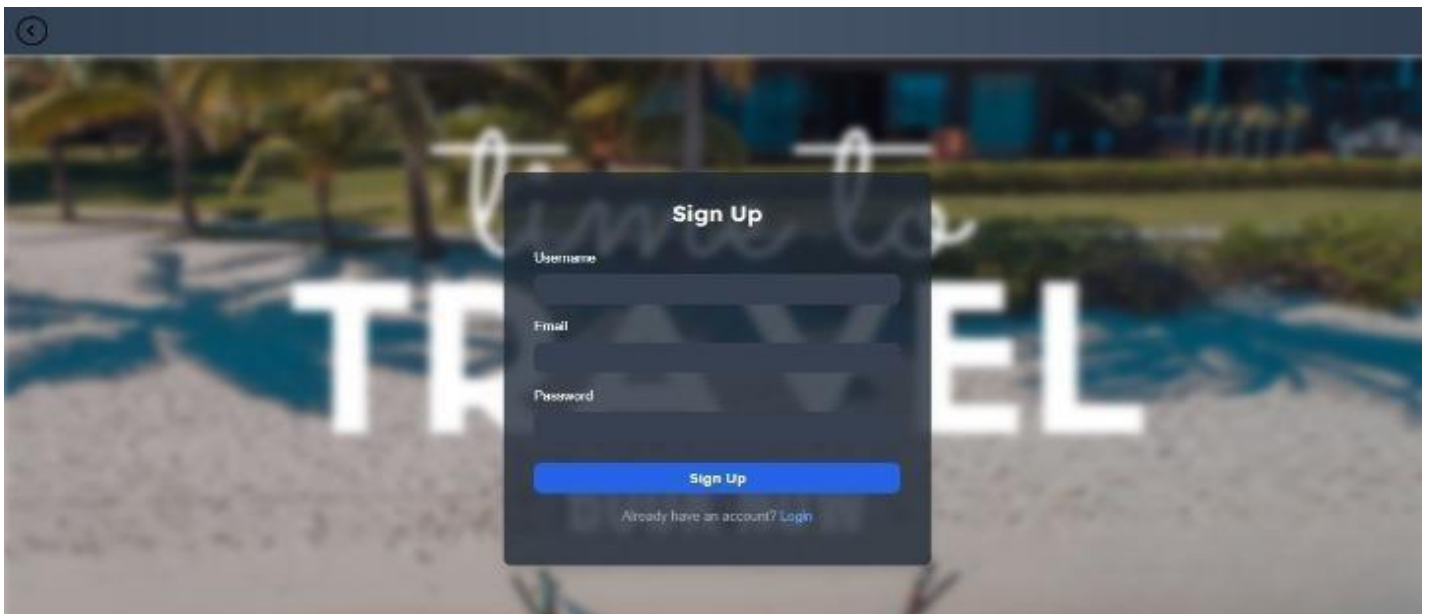
Password  
\*\*\*\*\*

## Vistelligence

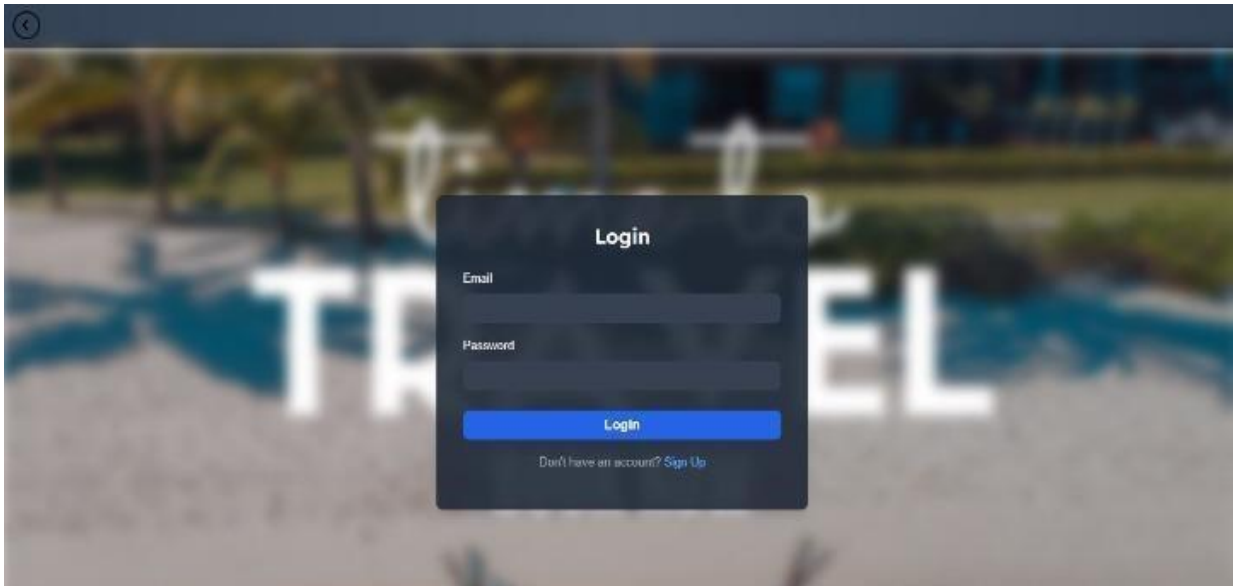
### Profile :



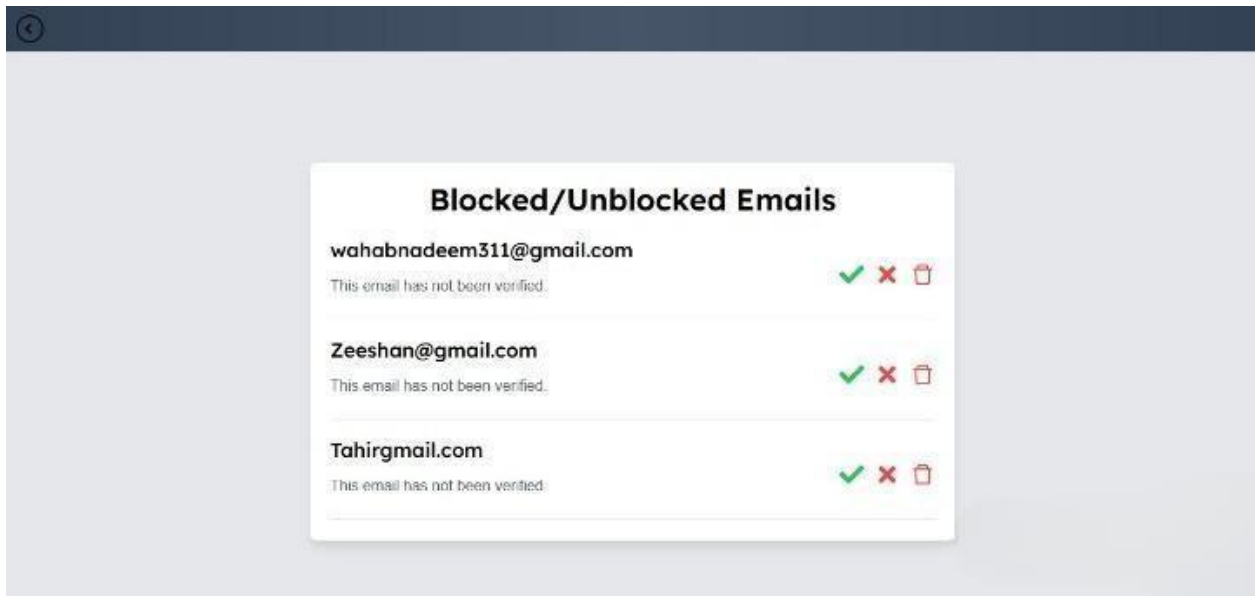
### Sign-Up Page:



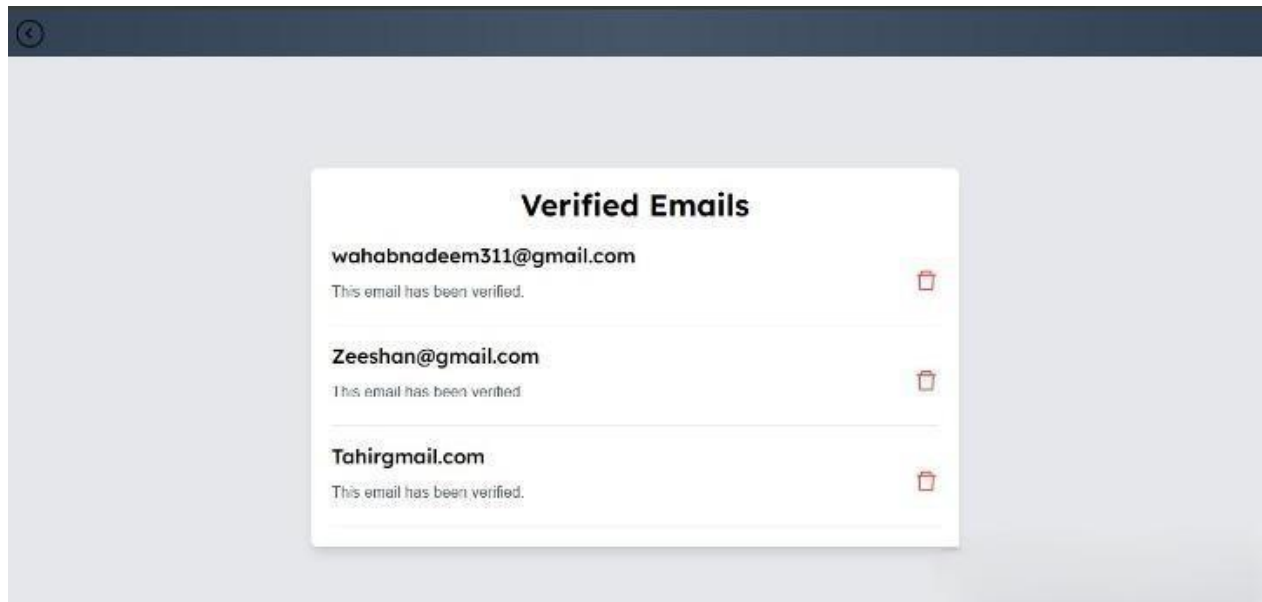
### Login-In Page:



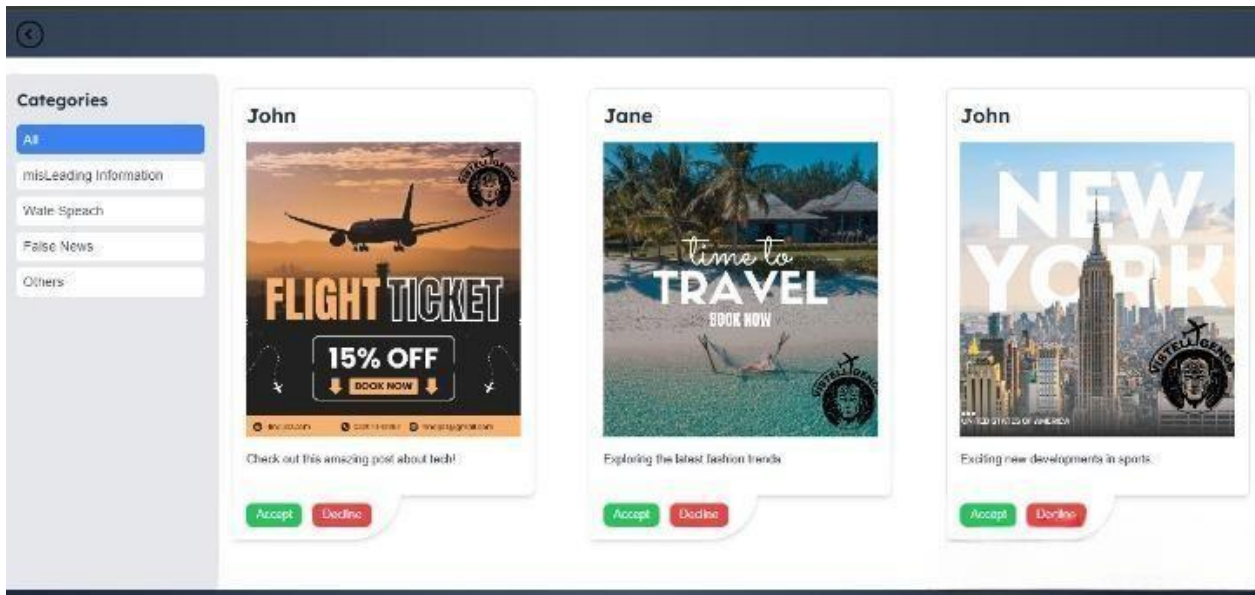
### Block/Unblock Email:



## Verified Emails:



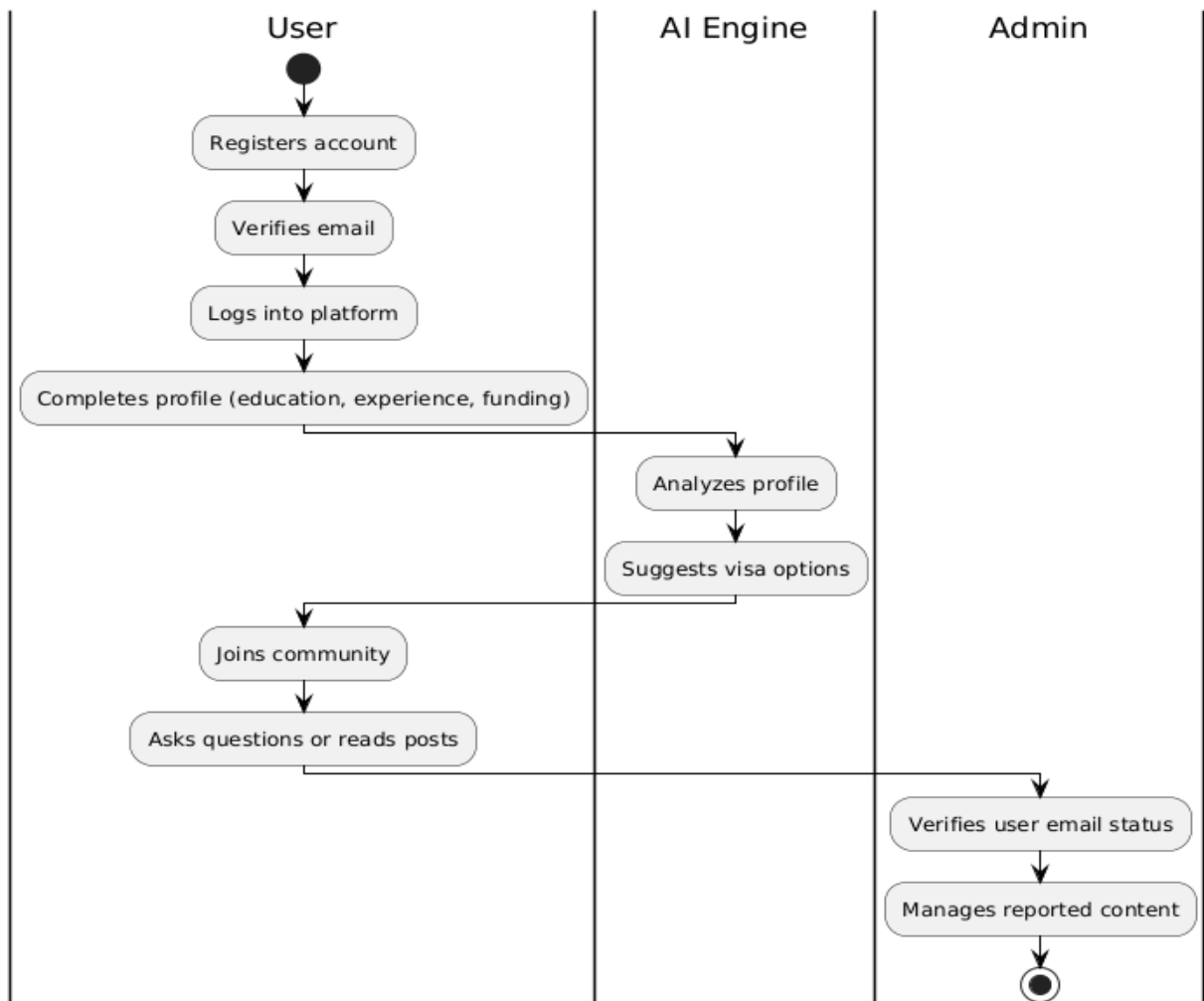
## Reported-Post:



## 4.1 Workflow

### Complete Workflow:

1. User registers and verifies email.
2. User logs into the platform.
3. Completes profile with education, experience, and funding info.
4. AI Engine analyzes profile and suggests visa options.
5. User can join the community, ask questions, or read posts.
6. Admin manages user-reported content and verifies email statuses.



**Figure 4.1: Swimlane daigram**

*For further details, review Figure 2.2.5:Swimlane diagram*

## 4.2 Screens

1. **Signup Screen:** User registers with email, password, and phone number.
2. **Login Screen:** User login with verified credentials.
3. **Dashboard:** Displays profile, visa recommendations, and community access.
4. **Create Post Page:** User can post questions or experiences.
5. **Community Feed:** List of user posts with reply options.
6. **Admin Panel:** View, block/unblock/delete users and posts.

## 4.3 Additional Information

*<NO additional interfaces beyond what has been described.>*

# 5. Other Design Details

## 5.1 AI Model Details

- **Features Considered:** Education Level, Experience, Financial Resources, Country Preference.
- **Data set:**  
<https://docs.google.com/document/d/1ybK7a10n1Fgk49Poj4FK8yfZOBcFF8SxkX-oN0fHWso/edit?usp=drivesdk>

**Note :** “Our model is Rag-Based”

## 5.2 Security Details

- **Two-Factor Authentication:** Optional feature planned.

## 5.3 Design Principles

- **Single Responsibility Principle:** Each module has one clearly defined purpose.
- **Separation of Concerns:** Frontend and backend are strictly separated.
- **DRY (Don't Repeat Yourself):** Reusable React components and Node.js middleware used.

# 6. Test Specification and Results

## 6.1 Test Case Specification

**Table 6.1: TC-1: User registration**

|            |      |
|------------|------|
| Identifier | TC-1 |
|------------|------|



|                                |                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Related requirements(s)</b> | UC-01: User Registration (SRS Section 3.13, Page 13)                                                                                        |
| <b>Short description</b>       | Test valid user registration with correct input data                                                                                        |
| <b>Pre-condition(s)</b>        | User is on the registration page; not logged in                                                                                             |
| <b>Input data</b>              | Name: "tester12"<br>Email: "tester1234@gmail.com"<br>Password: "Tester707@"<br>Confirmed password: "Tester707@"<br>Phone Number:03449090112 |
| <b>Detailed steps</b>          | 1. Navigate the page<br>2. Enter the credentials (input data)                                                                               |
| <b>Expected result(s)</b>      | User is successfully registered and redirected to login page                                                                                |
| <b>Post-condition(s)</b>       | The user <b>account is not created</b> due to input validation failure. The system <b>does not store the provided registration data</b>     |
| <b>Actual result(s)</b>        | User is not successfully registered                                                                                                         |
| <b>Test Case Result</b>        | <b>Fail</b> – UI does not meet expected input structure as per requirements.                                                                |

Table 6.2: TC-2: User registration

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-2                                                                                                    |
| <b>Related requirements(s)</b> | UC-01: User Registration (SRS Section 3.13, Page 13)                                                    |
| <b>Short description</b>       | Test valid user registration with correct input data                                                    |
| <b>Pre-condition(s)</b>        | User is on the registration page; not logged in                                                         |
| <b>Input data</b>              | Name: "tester12"<br>Email: "tester1234@gmail.com"<br>Password: "Tester707@"<br>Phone Number:03449090112 |
| <b>Detailed steps</b>          | 1. Navigate the page<br>2. Enter the credentials (input data)<br>3. Click the "Sign-up button"          |
| <b>Expected result(s)</b>      |                                                                                                         |

|                          |                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
|                          | User is successfully registered and redirected to login page                                                            |
| <b>Post-condition(s)</b> | A new user account is successfully created and stored in the system database. The user is redirected to the login page. |
| <b>Actual result(s)</b>  | User is successfully registered and redirected to login page                                                            |
| <b>Test Case Result</b>  | Pass                                                                                                                    |

Table 6.3: TC-3: User registration

|                                |                                                                                                                                                                                                                  |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-3                                                                                                                                                                                                             |
| <b>Related requirements(s)</b> | UC-01: User Registration (SRS Section 3.13, Page 13)                                                                                                                                                             |
| <b>Short description</b>       | Test valid user registration with correct input data                                                                                                                                                             |
| <b>Pre-condition(s)</b>        | User is on the registration page; not logged in                                                                                                                                                                  |
| <b>Input data</b>              | Name: "tester 2"<br>Email: "tester1234@gmail.com"<br>Password: "Xyz@123"<br>Phone Number: "0326155054"                                                                                                           |
| <b>Detailed steps</b>          | 1. Navigate the page<br>2. Enter the credentials (input data)<br>3. Click the "Sign-up button"                                                                                                                   |
| <b>Expected result(s)</b>      | User shouldn't successfully registered                                                                                                                                                                           |
| <b>Post-condition(s)</b>       | The user account is <b>not created</b> because the provided email address already exists in the system. The registration process is halted, and the user remains on the registration page with an error message. |
| <b>Actual result(s)</b>        | Registration is failed because the registration email is already exist                                                                                                                                           |
| <b>Test Case Result</b>        | Pass                                                                                                                                                                                                             |

Table 6.4: TC-4: User login

|                                |                                                         |
|--------------------------------|---------------------------------------------------------|
| <b>Identifier</b>              | TC-4                                                    |
| <b>Related requirements(s)</b> | UC-01: User Login (SRS Section 3.13, Page 13)           |
| <b>Short description</b>       | Test login successfully with correct input data         |
| <b>Pre-condition(s)</b>        | User should be registered; not logged in                |
| <b>Input data</b>              | Email: "tester1234@gmail.com"<br>Password: "Tester707@" |

|                           |                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           |                                                                                                                                                                |
| <b>Detailed steps</b>     | <ol style="list-style-type: none"> <li>1. Navigate the page</li> <li>2. Enter the credentials (input data)</li> <li>3. Click the “login button”</li> </ol>     |
| <b>Expected result(s)</b> | User is successfully login and redirected to Vistelligence (AI model page)                                                                                     |
| <b>Post-condition(s)</b>  | The user is authenticated, an active session is initiated, and the user is redirected to the Vistelligence (AI model page) with access to authorized features. |
| <b>Actual result(s)</b>   | User is successfully login and redirected to Vistelligence (AI model page)                                                                                     |
| <b>Test Case Result</b>   | Pass                                                                                                                                                           |

Table 6.5: TC-5: User authentication

|                                |                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-5                                                                                                                                                                                     |
| <b>Related requirements(s)</b> | UC-01: User Authentication (SRS Section 3.13, Page 13)                                                                                                                                   |
| <b>Short description</b>       | Test valid user registration with correct input data                                                                                                                                     |
| <b>Pre-condition(s)</b>        | User should be registered; not logged in                                                                                                                                                 |
| <b>Input data</b>              | Email: "tester1234@gmail.com"                                                                                                                                                            |
| <b>Detailed steps</b>          | <ol style="list-style-type: none"> <li>1. Navigate the page</li> <li>2. Enter the credentials (input data)</li> <li>3. Click on the link of “forget password”</li> </ol>                 |
| <b>Expected result(s)</b>      | Navigate to the forget password field                                                                                                                                                    |
| <b>Post-condition(s)</b>       | The system redirects the user to the "Forget Password" interface where the user can initiate the password recovery process (e.g., by entering a recovery email to receive a reset link). |
| <b>Actual result(s)</b>        | Navigate to the forget password field                                                                                                                                                    |
| <b>Test Case Result</b>        | pass                                                                                                                                                                                     |

Table 6.6: TC-6: User authentication

|                                |                                                        |
|--------------------------------|--------------------------------------------------------|
| <b>Identifier</b>              | TC-6                                                   |
| <b>Related requirements(s)</b> | UC-01: User Authentication (SRS Section 3.13, Page 13) |
| <b>Short description</b>       | Verify valid user registration with correct input data |

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-condition(s)</b>   | User should be registered; not logged in                                                                                        |
| <b>Input data</b>         | Email: "tester1234@gmail.com"                                                                                                   |
| <b>Detailed steps</b>     | 1. Navigate the page<br>2. Enter the credentials (input data)<br>3. Click on the link of “sent reset link button”               |
| <b>Expected result(s)</b> | Link has been sent to the user’s provided email                                                                                 |
| <b>Post-condition(s)</b>  | The password reset link <b>is not sent</b> due to a system error. The user remains on the reset page and sees an error message. |
| <b>Actual result(s)</b>   | Showing error message                                                                                                           |
| <b>Test Case Result</b>   | fail                                                                                                                            |

Table 6.7: TC-7: User Dashboard/Profile Module

|                                |                                                                                                                                                                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-7                                                                                                                                                                                                                                                |
| <b>Related requirements(s)</b> | UC-01: User Profile Management (SRS Section 3.13, Page 13)<br>SRS Section 2.2: User Classes and Characteristics (Profile Creation and Management)<br>SRS Section 4.3: Security Requirements (Input Validation)                                      |
| <b>Short description</b>       | User can update the profile                                                                                                                                                                                                                         |
| <b>Pre-condition(s)</b>        | User should be registered and logged-in                                                                                                                                                                                                             |
| <b>Input data</b>              | Enter the Name,Email,phone No                                                                                                                                                                                                                       |
| <b>Detailed steps</b>          | 1. Navigate to the Profile Page<br>2. Click to the setting button<br>3. Click to the update button<br>4. Click on the required filed which need to be updated<br>5. Enter the new information into the field<br>6. Click on the save changes button |
| <b>Expected result(s)</b>      | The changes has been saved                                                                                                                                                                                                                          |

|                          |                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Post-condition(s)</b> | The user's profile information is updated in the MongoDB database. The user remains on the profile page with the updated information displayed. |
| <b>Actual result(s)</b>  | System saves the changes and displays a confirmation message                                                                                    |
| <b>Test Case Result</b>  | Pass                                                                                                                                            |

**Table 6.8: TC-8: Admin Pannel**

|                                |                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-8:                                                                                                                                                                                    |
| <b>Related requirements(s)</b> | UC-4: Admin Login (SRS Section 3.19, Page 19)<br>SRS Section 2.2: User Classes and Characteristics (Admin Responsibilities)<br>SRS Section 4.3: Security Requirements (Input Validation) |
| <b>Short description</b>       | Verify that the admin can log in with valid credentials                                                                                                                                  |
| <b>Pre-condition(s)</b>        | Admin account must exist in the system                                                                                                                                                   |
| <b>Input data</b>              | Email: admin@vistelligent.com<br>Password: Admin@123                                                                                                                                     |
| <b>Detailed steps</b>          | 1. Navigate to the admin-login<br>2. Enter valid email and password<br>3. Click "Login" button                                                                                           |
| <b>Expected result(s)</b>      | System authenticates the admin, and the admin dashboard is displayed                                                                                                                     |
| <b>Post-condition(s)</b>       | Admin session is active                                                                                                                                                                  |
| <b>Actual result(s)</b>        | Admin dashboard is displayed                                                                                                                                                             |
| <b>Test Case Result</b>        | Pass                                                                                                                                                                                     |

**Table 6.09: TC-09 : Verify Display of reported Email with Verification Status**

|                                |                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-9 :                                                                                                                                                                                                          |
| <b>Related requirements(s)</b> | UC-6: View User Reports (SRS Section 3.20, Page 20)<br>SRS Section 2.2: User Classes and Characteristics (Admin Responsibilities: User Management)<br>SRS Section 4.3: Security Requirements (Input Validation) |

|                           |                                                                                                                                                                                |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Short description</b>  | Ensure that the system displays the verification status of each email                                                                                                          |
| <b>Pre-condition(s)</b>   | Admin is logged in and on the "Blocked/Unblocked Emails" page                                                                                                                  |
| <b>Input data</b>         | List of registered users with mixed verified/unverified email statuses                                                                                                         |
| <b>Detailed steps</b>     | <ol style="list-style-type: none"> <li>1. Navigate to the admin panel.</li> <li>2. Open the "Blocked/Unblocked Emails" section.</li> <li>3. Observe the email list.</li> </ol> |
| <b>Expected result(s)</b> | System displays a list of emails with their verification status (e.g., "Verified" or "This email has not been verified")                                                       |
| <b>Post-condition(s)</b>  | Email verification status is visible to admin                                                                                                                                  |
| <b>Actual result(s)</b>   | System displays the email list with correct verification statuses                                                                                                              |
| <b>Test Case Result</b>   | Pass                                                                                                                                                                           |

Table 6.10: TC-10 Block the reported Email

|                                |                                                                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-10:                                                                                                                                                  |
| <b>Related requirements(s)</b> | <b>SRS 2.2:</b> Admin manages user roles and access to visa data.<br><b>SRS 4.3:</b> Validate input data for security and system integrity              |
| <b>Short description</b>       | Verify that an email can be blocked by clicking the icon                                                                                                |
| <b>Pre-condition(s)</b>        | Admin is on the blocked /unblocked email list page.                                                                                                     |
| <b>Input data</b>              | Reported email "test_unverified@gmail.com"                                                                                                              |
| <b>Detailed steps</b>          | <ol style="list-style-type: none"> <li>1. Navigate to the list of "Blocked/Unblocked Emails" page</li> <li>2. Confirm the action if prompted</li> </ol> |
| <b>Expected result(s)</b>      | Email status changes to blocked, icon feedback updates                                                                                                  |
| <b>Post-condition(s)</b>       | Email is marked as blocked                                                                                                                              |
| <b>Actual result(s)</b>        | Email status changes to "Blocked"                                                                                                                       |
| <b>Test Case Result</b>        | Pass                                                                                                                                                    |

Table 6.11:TC-11: Unblock the reported Email

|                                |                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-11:                                                                                                                                     |
| <b>Related requirements(s)</b> | <b>SRS 2.2:</b> Admin manages user roles and access to visa data.<br><b>SRS 4.3:</b> Validate input data for security and system integrity |
| <b>Short description</b>       | Verify that an email can be blocked by clicking the icon                                                                                   |
| <b>Pre-condition(s)</b>        | Admin is on the blocked /unblocked email list page.                                                                                        |

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|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <b>Input data</b>         | Reported email “”                                                      |
| <b>Detailed steps</b>     | 1. Navigate to the list of emails<br>2. Confirm the action if prompted |
| <b>Expected result(s)</b> | Email status changes to unblocked, icon feedback updates               |
| <b>Post-condition(s)</b>  | Email is marked as unblocked                                           |
| <b>Actual result(s)</b>   | Email status changes to “Unblocked”                                    |
| <b>Test Case Result</b>   | Pass                                                                   |

**Table 6.12: TC-12: Delete the reported Email from Block/unblocked page**

|                                |                                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-12:                                                                                                                                     |
| <b>Related requirements(s)</b> | <b>SRS 2.2:</b> Admin manages user roles and access to visa data.<br><b>SRS 4.3:</b> Validate input data for security and system integrity |
| <b>Short description</b>       | Verify the delete functionality for emails.                                                                                                |
| <b>Pre-condition(s)</b>        | Admin has access to user emails.                                                                                                           |
| <b>Input data</b>              | Email to be deleted                                                                                                                        |
| <b>Detailed steps</b>          | 1. Click the icon of delete next to an email<br>2. Confirm deletion if needed                                                              |
| <b>Expected result(s)</b>      | Email is removed from the system                                                                                                           |
| <b>Post-condition(s)</b>       | Email is permanently deleted                                                                                                               |
| <b>Actual result(s)</b>        | Email is deleted                                                                                                                           |
| <b>Test Case Result</b>        | Pass                                                                                                                                       |

**Table 6.13: TC-13: Visa Option Search**

|                                |                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-13:                                                                                                                             |
| <b>Related requirements(s)</b> | UC-1: Search for Visa Options (SRS Section 3.14, Page 14)<br>SRS Section 2.1: Product Features (Personalized Visa Recommendations) |

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|                           |                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Short description</b>  | Verify that the system provides accurate visa options based on user profile.                                                                                    |
| <b>Pre-condition(s)</b>   | User has provided education, experience, and funding details.                                                                                                   |
| <b>Input data</b>         | Education: Bachelor's degree<br><br>Experience: 1 year<br><br>Available Funds: \$20,000<br><br>Target Countries: Canada, Australia, Malaysia, US, UK            |
| <b>Detailed steps</b>     | 1. User submits personal profile through the UI.<br>2. System processes input and queries visa options.<br>3. System generates visa suggestions and likelihood. |
| <b>Expected result(s)</b> | System suggests visa types and likelihood of approval per country.<br>E.g., “Canada: Moderate to High,” “Malaysia: High,” etc.                                  |
| <b>Post-condition(s)</b>  | System stores or displays response.                                                                                                                             |
| <b>Actual result(s)</b>   |                                                                                                                                                                 |
| <b>Test Case Result</b>   |                                                                                                                                                                 |

**Table 6.14: TC-14: Visa Approval Prediction:**

|                                |                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-14:                                                                                                                                               |
| <b>Related requirements(s)</b> | UC-2: Predict Visa Approval Chances (SRS Section 3.16, Page 16)<br><br>SRS Section 2.1: Product Features (Visa Approval Predictions)                 |
| <b>Short description</b>       | Ensure the system predicts visa approval chances based on profile.                                                                                   |
| <b>Pre-condition(s)</b>        | Visa options already generated.                                                                                                                      |
| <b>Input data</b>              | Education: Bachelor's degree<br><br>Experience: 1 year<br><br>Available Funds: \$20,000<br><br>Target Countries: Canada, Australia, Malaysia, US, UK |
| <b>Detailed steps</b>          | 1. System analyzes user's qualifications and funding.                                                                                                |



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|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
|                           | 2. System assigns a percentage or qualitative rating for each country. |
| <b>Expected result(s)</b> | E.g., “Visa Approval Chance: 85% for US” or “Malaysia: High”           |
| <b>Post-condition(s)</b>  | Approval prediction stored in report.                                  |
| <b>Actual result(s)</b>   |                                                                        |
| <b>Test Case Result</b>   |                                                                        |

**Table 6.15: TC-15: Visa Interview Preparation;**

|                                |                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-15:                                                                                                                                                   |
| <b>Related requirements(s)</b> | UC-3: Prepare for Visa Interview (SRS Section 3.17, Page 17)<br>SRS Section 2.1: Product Features (Interview Preparation)                                |
| <b>Short description</b>       | Test whether interview questions and suggestions are tailored to profile.                                                                                |
| <b>Pre-condition(s)</b>        | Visa type selected.                                                                                                                                      |
| <b>Input data</b>              | Country: United States<br><br>Profile: Same as above                                                                                                     |
| <b>Detailed steps</b>          | 1. User requests interview preparation help.<br>2. System returns sample questions and prep tips.                                                        |
| <b>Expected result(s)</b>      | Questions like “Why this university?” and “How will you finance your studies?”<br><br>Custom prep tips provided (e.g., ties to home country, SOP advice) |
| <b>Post-condition(s)</b>       | Info stored or shown to user.                                                                                                                            |
| <b>Actual result(s)</b>        |                                                                                                                                                          |
| <b>Test Case Result</b>        |                                                                                                                                                          |

**Table 6.16: TC-16: Cost Estimation;**

|                                |                                                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-16:                                                                                                                          |
| <b>Related requirements(s)</b> | SRS Section 2.1: Product Features (Real-Time Cost Estimations)<br>SRS Section 1.3: Scope (Comprehensive Application Strategies) |

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|                           |                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Short description</b>  | Ensure system returns detailed visa and travel cost estimation per country.                                         |
| <b>Pre-condition(s)</b>   | Country selection is made.                                                                                          |
| <b>Input data</b>         | Selected Countries: US, UK                                                                                          |
| <b>Detailed steps</b>     | 1. User requests cost estimation.<br>2. System returns visa fee, travel costs, processing fees, and other expenses. |
| <b>Expected result(s)</b> | US: ~\$833<br><br>UK: ~\$950<br><br>All cost components itemized                                                    |
| <b>Post-condition(s)</b>  | Cost breakdown shown or stored.                                                                                     |
| <b>Actual result(s)</b>   |                                                                                                                     |
| <b>Test Case Result</b>   |                                                                                                                     |

**Table 6.17: TC-17:**

|                                |                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-17:                                                                                                                                                                                               |
| <b>Related requirements(s)</b> | UC-1: Search for Visa Options (SRS Section 3.14, Page 14)<br>UC-2: Predict Visa Approval Chances (SRS Section 3.16, Page 16)<br><br>SRS Section 2.1: Product Features (Personalized Recommendations) |
| <b>Short description</b>       | User enters <b>very low funds (\$5,000)</b> to test financial threshold impact.                                                                                                                      |
| <b>Pre-condition(s)</b>        | Profile submission screen active                                                                                                                                                                     |
| <b>Input data</b>              | Education: Bachelor's degree<br><br>Experience: 2 years<br><br>Funds: \$5,000<br><br>Countries: US, UK, Malaysia                                                                                     |
| <b>Detailed steps</b>          | 1. Submit profile.<br>2. System calculates feasibility and chances.                                                                                                                                  |
| <b>Expected result(s)</b>      | US and UK: "Low" or "Challenging" approval                                                                                                                                                           |

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|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
|                          | Malaysia: Possibly “Moderate” or warning about insufficient funds |
| <b>Post-condition(s)</b> | System warns about fund shortfall                                 |
| <b>Actual result(s)</b>  |                                                                   |
| <b>Test Case Result</b>  |                                                                   |

**Table 6.18: TC-18: Vague SOP Request**

|                                |                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-18:                                                                                                                                                                                 |
| <b>Related requirements(s)</b> | UC-3: Prepare for Visa Interview (SRS Section 3.17, Page 17)<br>SRS Section 2.1: Product Features (Interview Preparation)<br>SRS Section 4.3: Security Requirements (Input Validation) |
| <b>Short description</b>       | User enters an <b>unclear or vague SOP</b> request for interview prep.                                                                                                                 |
| <b>Pre-condition(s)</b>        | Interview prep module active                                                                                                                                                           |
| <b>Input data</b>              | SOP: “I want to go abroad to study anything good.”                                                                                                                                     |
| <b>Detailed steps</b>          | 1. Submit SOP for interview preparation tips.<br>2. System parses and analyzes clarity.                                                                                                |
| <b>Expected result(s)</b>      | System suggests rewriting SOP<br><br>Prompts user to provide academic goals and target programs                                                                                        |
| <b>Post-condition(s)</b>       | SOP feedback provided                                                                                                                                                                  |
| <b>Actual result(s)</b>        |                                                                                                                                                                                        |
| <b>Test Case Result</b>        |                                                                                                                                                                                        |

**Table 6.19: TC-19:**

|                                |                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-19:                                                                                                                           |
| <b>Related requirements(s)</b> | UC-2: Predict Visa Approval Chances (SRS Section 3.16, Page 16)<br>SRS Section 2.1: Product Features (Visa Approval Predictions) |
| <b>Short description</b>       | Cost estimation for <b>a rare or unsupported country</b> .                                                                       |
| <b>Pre-condition(s)</b>        | User selects a non-default country                                                                                               |
| <b>Input data</b>              | Country: Latvia                                                                                                                  |
| <b>Detailed steps</b>          | 1. Request visa cost breakdown<br>2. System processes unsupported country input                                                  |

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|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| <b>Expected result(s)</b> | “Country not supported” or fallback to general cost ranges |
| <b>Post-condition(s)</b>  | Error message or default fallback                          |
| <b>Actual result(s)</b>   |                                                            |
| <b>Test Case Result</b>   |                                                            |

**Table 6.20: TC-20:**

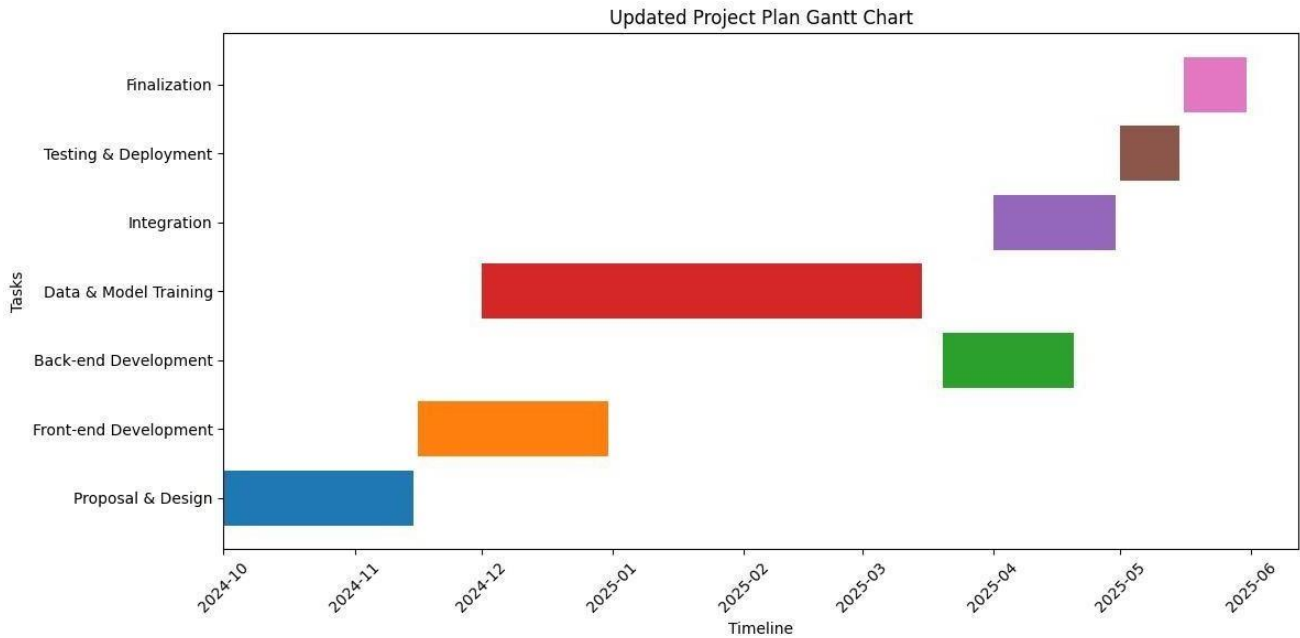
|                                |                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Identifier</b>              | TC-20:                                                                                                                           |
| <b>Related requirements(s)</b> | UC-2: Predict Visa Approval Chances (SRS Section 3.16, Page 16)<br>SRS Section 2.1: Product Features (Visa Approval Predictions) |
| <b>Short description</b>       | Profile with <b>PhD and full scholarship</b> to test high visa success likelihood                                                |
| <b>Pre-condition(s)</b>        | High-achievement profile submitted                                                                                               |
| <b>Input data</b>              | Education: PhD in Computer Science, Experience: 5 years research, Funds: Full scholarship, Country: UK                           |
| <b>Detailed steps</b>          | 1. Submit high-profile data<br>2. System returns visa prediction                                                                 |
| <b>Expected result(s)</b>      | Visa Approval Prediction: “Very High” or 95%                                                                                     |
| <b>Post-condition(s)</b>       | Prediction saved                                                                                                                 |
| <b>Actual result(s)</b>        |                                                                                                                                  |
| <b>Test Case Result</b>        |                                                                                                                                  |

## 6.2 Summary of Test Results

**Table 6.2: Summary of Test Results**

| Module Name                        | Test cases run                     | Number of defects found | Number of defects corrected so far | Number of defects still need to be corrected |
|------------------------------------|------------------------------------|-------------------------|------------------------------------|----------------------------------------------|
| User Registration & Authentication | TC-1, TC-2, TC-3, TC-4, TC-5, TC-6 | 2                       | 0                                  | 2                                            |
| User Profile Module                | TC-7                               | 0                       | 0                                  | 0                                            |
| Admin Panel                        | TC-8, TC-9, TC-10, TC-11, TC-12    | 0                       | 0                                  | 0                                            |
| Visa Option Search                 | TC-13                              |                         |                                    |                                              |
| Visa Approval Prediction           | TC-14, TC-17, TC-19, TC-20         |                         |                                    |                                              |
| Visa Interview Preparation         | TC-15, TC-18                       |                         |                                    |                                              |
| Cost Estimation                    | TC-16                              |                         |                                    |                                              |
|                                    |                                    |                         |                                    |                                              |
| <b>Complete System</b>             | TC-1 to TC-20                      |                         | 0                                  | 2                                            |

## 7. Revised Project Plan



**Table 7.1 : Project Completion Status**

| Module Name                        | Status<br>(Complete, Partially Implemented, Not Implemented) |
|------------------------------------|--------------------------------------------------------------|
| User Registration & Authentication | Complete                                                     |
| User Profile Management            | Complete                                                     |
| AI Recommendation Engine           | Partially Completed                                          |
| Community Forum                    | Complete                                                     |
| Admin Panel                        | Complete                                                     |
| Cost Estimator                     | Partially Completed                                          |
| Complete System                    | Partially Completed                                          |

## 8. References

React Documentation: <https://react.dev/>

Node.js Documentation: <https://nodejs.org/en/docs/>

MongoDB Documentation: <https://www.mongodb.com/docs/>

Bootstrap Documentation: <https://getbootstrap.com/docs/>

AWS Documentation: <https://aws.amazon.com/documentation/>

Data set : <https://docs.google.com/document/d/1ybK7a10n1Fgk49Poj4FK8yfZOBcFF8SxkX-oN0fHWso/edit?usp=drivesdk>

### Appendix A: Glossary

| Term | Description                       |
|------|-----------------------------------|
| AI   | Artificial Intelligence           |
| JWT  | JSON Web Token                    |
| UI   | User Interface                    |
| UX   | User Experience                   |
| API  | Application Programming Interface |
| SMTP | Simple Mail Transfer Protocol     |

## Appendix B: IV & V Report

(Independent verification & validation)

IV & V Resource

Name

Signature

| S# | Defect Description                     | Origin Stage        | Status  | Fix Time |         |
|----|----------------------------------------|---------------------|---------|----------|---------|
|    |                                        |                     |         | Hours    | Minutes |
| 1  | Password validation missing            | Development         | Pending |          |         |
| 2  | Forget password email failure          | Testing             | Pending |          |         |
| 3  | UI feedback missing on admin actions   | Testing             | Pending |          |         |
| 4  | Estimation fallback for rare countries | Integration Testing | Pending |          |         |

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

**Table 1: List of non-trivial defects**

*The system was independently verified against the SRS, SDS, and design documents.  
All critical functional requirements were validated through manual testing.  
All major defects identified will be fixed before the final system build.*

This document has been adapted from the following:

- Previous project templates at UCP
- High-level Technical Design, Centers for Medicare & Medicaid Services. ([www.cms.gov](http://www.cms.gov))